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# Design and Fabrication of an 8-point UAV Testing Rig

# FYP-20-08

## Interim Report

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October 5, 2024

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# Declaration

We hereby declare that the work contained in this report is original; researched and documented by the undersigned students. It has not been used or presented elsewhere in any form for award of any academic qualification or otherwise. Any material obtained from other parties have been duly acknowledged. We have ensured that no violation of copyright or intellectual property rights have been committed.

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# Abstract

Unmanned Aerial Vehicles are aircrafts that can be piloted remotely to perform certain tasks. Testing of these Unmanned Aerial Vehicles remains crucial to ensure safety in operation and the accuracy of processing the commands given to the UAV by its user. Several methods of testing UAVs have been developed such as the Gimbal Test Rig and the Column Test Stand. However, the testing methods are relatively expensive to acquire, with locally manufactured options being unavailable which makes it difficult for local UAV manufacturers to compete in the global market. This project aims to develop a locally available Unmanned Aerial Vehicle testing rig for the measurement of desired UAV parameters. The mechanical structure, the electrical circuit and the control system of the test rig have been designed. The next phase of the study is the fabrication and testing of the test bed. It is expected that the test bed will be able to accurately measure set UAV parameters for tests conducted on it. The rig is aimed at providing a budget-friendly option for the testing that will be available locally for those venturing into the study of UAVs and general enthusiasts.

# Contents

# List of Figures

# List of Tables

# List of Abbreviations

| | |
|---|---|
| BLE | Bluetooth Low Energy |
| CFD | Computational Fluid Analysis |
| CSV | Comma Separated values |
| DOF | Degree of Freedom |
| FEM | Finite Element Method |
| JSON | Javascript Object Notation |
| UAV | Unmanned Aerial Vehicle |
| VTOL | Vertical Take-off and Landing |

# 1 Introduction

## 1.1 Background

The first Unmanned Aerial Vehicle (UAV), developed by Joseph-Michel and Jacques-Etienne Montgolfier, was a hot-air balloon as this does not require a human to pilot [1]. The balloon was modified to carry bombs and used by lieutenant Franz Von Uchatius in 1849 to attack Venice. Later on, other uses were found for UAVs such as aerial photography in 1896, surveillance in 1973, personal use in 2006 and delivery in non-accessible areas in 2013 [1]. The latter was highly adopted by major companies such as UPS, Amazon and Google. UAV usage has been rapidly expanding ever since and was widely adopted during the COVID-19 pandemic in 2020 for the delivery of medical supplies [1].

UAV technology has expanded in terms of size and capability, with the era of miniaturization technology allowing developers to create increasingly smaller UAVs able to fit in on one's palm such as the DJI Mini 2 SE that has overall dimensions of $138 \times 81 \times 58$ mm [2]. Larger UAVs have greater endurance and payload capacity and have helped revolutionize industries such as the agricultural and surveillance industries in Kenya. KenDrone is an example of a UAV company founded in Kenya that provides UAV services including film, thermal inspection, security and training [3].

With the increase in the use of UAVs came the need to test them. The reasons for the testing of UAVs are the need to ensure reliability of the UAVs made by manufacturers, the safety of consumers and for research in various institutions. The increase in regulations set by the authorities on the making and use of UAVs has also driven the design and capabilities of these aircraft [3].

The parameters tested in UAVs range from the mechanical attributes of the UAV such as the thrust of the UAV and the structural integrity of the body of the UAV to the electrical and control features such as the power requirements of the UAV, the accuracy and reliability of the sensors and the communication strength of the UAV [4].

Several methods have been developed for the testing of UAVs. These include the open-field tests, the use of UAV testing rigs and the use of simulations. Open-field tests involve the flying of UAVs in the open environment and is preferred for testing parameters such as communication range that cannot be easily modelled. Simulations involve the building of models of both the UAV and its expected environment and observing their interactions that can either be a physical representation of the system to be tested or a computer model that is taken through analyses of various parameters. They are mostly used in the design phase for optimization purposes or for tests that may be too dangerous or costly to conduct using the other methods [4]. Testing rigs involve constraining the UAV, in part or whole, for the tests and is preferred when the UAV test is to be conducted in a controlled environment.

Test beds are specialized equipment that allow intricate measurement of desired UAV parameters such as motor and propeller performance for lift and movement that would aid in ensuring the reliability of UAVs if they are in the hands of consumers and encourage companies to implement this technology in their everyday operations to make work easier. These methods have been implemented by each manufacturer for testing and commissioning of their UAVs.

## 1.2 Problem statement

Increase in the use of UAVs has come with the increase in the need to test them reliably. UAV implementation in Kenya has allowed industrial revolution such as transportation in agricultural farms and surveillance in the security industry, to creating our own UAVs. This increase has led to the need for different means of testing the reliability of UAVs, among them being the use of testing rigs.

In spite of the development of different UAV testing rigs, these rigs remain specialized and expensive as most companies develop rigs for in-house testing of their UAVs as most of them are catered for specific UAVs leaving them rather inflexible. A need, therefore, arises to design and fabricate a reliable testing rig for the research and development of UAVs

## 1.3 Objectives

### Main objective

The main objective of the study is design and fabricate an experimental platform for the measurement and testing of UAV thrust parameters.

### Specific objectives

1. To design and fabricate a UAV testing rig that will facilitate UAV parameter measurement
2. To develop a control system that will control the collection and processing of data from the rig.
3. To calibrate and test the UAV testing rig for accuracy and repeatability

## 1.4 Expected outcomes

The following are the expected outcomes of the UAV testing rig:

1. Successful fabrication of the UAV measurement system
2. Seamless recording and processing of data obtained from the UAV testing rig
3. Successful calibration and testing of the rig for accuracy and repeatability

## 1.5 Justification of the study

Unmanned Aerial Vehicles are an emergent technology that has found use in areas such as surveillance, commercial deliveries and even in the medical field. With the rise of the implementation of UAV technology, there is need to test them for assurance in quality of production by the manufacturing company that meets regulatory standards set by the government. Consumers such as the government would profit from reliable UAV technology and implement further advances in the security of a nation keeping its people safe.

# 2 Literature Review

## 2.1 UAV overview

The UAV is an unmanned aerial vehicle that can be flown remotely. It may be broadly classified using the following criteria [5]:

1. Build: This mainly concerns the way in which the UAV gets its lift. This classification gives rises to two main types of UAVs [6]. Fixed-wing UAVs, shown on Figure 2.1, have builds similar to those of a typical aeroplane and generating lift from their wings and propellers. They are known as fixed wings UAVs since they have rigid wings as major feature of their structure [7]. UAVs of this type are mainly used in military applications.

Figure 2.1: Fixed wing UAV [8]

Rotor UAVs are the most common types of domestic UAVs. They generate their lift from the use of rotors in designed in their builds. They may further be classified according to the number of rotors in their design. Common types include: Single-rotor UAVs Figure 2.3 (a), quad-copters (four rotors) Figure 2.3(b) and the tri-copter UAVs (three rotors).

(a) (b)

Figure 2.3: Types of rotor UAVs (a) Single Rotor UAVs [9] (b) Quad-copter UAVs [10]

Hybrid UAVs are UAVs that have both rotors and fixed-wings in their design, also known as the VTOL UAVs. The UAVs under this classification use the rotors for take-off and landing and use the wings for flight [6].

2. Function: The functions of UAVs range from military uses (combat UAVs), delivery UAVs, surveillance UAVs, photography UAVs, toy UAVs and racing UAVs. The UAVs, classified by their functions all fall under the aforementioned categories, with changes made to their designs to suit the required functions. For instance photography UAVs are mainly multi-rotor UAVs due to their ability to hover over a place and the mapping UAVs used for geographical mapping that are mainly fixed wing UAVs due to their relative long flight times [11].

## 2.2 UAV Design and Development

With the increase in the manufacture of UAVs, there has arisen a need to reliably test them [12]. UAV technology is constantly changing and improving. Testing provides useful data on the underlying technology in UAVs in order to improve on the UAVs. Regulatory bodies also require UAV tests to ensure that they comply with the safety standards set. In addition to these, UAV tests are made to ensure safety for the consumer by ensuring that the UAVs meet safety thresholds. Testing also ensures that the design of the UAV is sufficient to carry out the function intended for it in the environment the UAV is to be used in.

UAV testing is the measurement of various parameters of the UAV and its parts e.g. communication range. UAV testing can be divided into three categories:

### 2.2.1 Open Field Testing

Open field tests are done mostly outdoors. These tests involve flying of the UAV in an uncontrolled environment while measuring the intended parameters. For instance, Cwiakala [13] used open field testing to test the accuracy of drones autonomous systems in comparison with the manufacturer's specifications, with the conclusion being that the standardization of these tests for application in various is required. The test is relatively easy to set-up and carry out and can provide a quick overview of the UAV [14]. These tests are suitable for measuring the behaviour of the UAV under different environmental conditions, the reliability of its location and communication systems, its landing and take-off behaviour, and other aspects of the UAV that can not be sufficiently measured in a closed environment or cannot be sufficiently be modelled for use in simulations [15]. Since the test may be relatively unsafe for both the UAV and its operator, extra precaution should be undertaken when conducting UAV tests in the open environment.

### 2.2.2 Closed environment testing

These tests involve constraining either a UAV model or the actual UAV to a closed environment system and carrying out the test. These tests are suitable for testing a particular system of the UAV e.g. propulsion system or the electrical system, especially where the environment for the test can easily be replicated. UAV Tests in controlled environments, specifically the wind tunnel, have been conducted by Forster et al. [16] to assess the accuracy of a flight dynamics simulation made from previous experimental data. While the simulations were accurate, they had to be updated to match the data found in the current tests. Closed environment testing is relatively safer than the open-air tests as everything is held down for the test. It, however, has the limitation not being able to completely replicate real world conditions.

Closed environment testing can be classified into wind tunnels and test rigs. Wind tunnels are used to simulate flight on an article in addition to showing the interaction of air with the model. Sensors are placed on the model to monitor its interaction with the airflow by measuring parameters such as pressure along the surface, lateral force and orientation of the model in the tunnel [17]. Test rigs are used for testing of the parts or whole UAVs in a closed system [18]. Some of the common testing rigs designs are the column test rigs, the gimbal test rigs and thrust stands.

A Column test rig is a 4 degree of freedom system where the UAV is mounted at its top for the test. The column is mounted onto its base by means of springs in addition to its base having a ball joint to ensure free movement of the UAV during testing [19]. The column system has the benefit of allowing for free movement of the UAV during the test. The system, however, is limited to the testing of whole UAVs as opposed to UAV parts. This is in addition to only having one point of measurement, making it relatively difficult to tell the distribution of forces in the UAV.

A gimbal system, Figure 2.5 (b), is a 3 degree of freedom system where the UAV is attached to a gimbal platform that can rotate to all orientations. The gimbal system has an inertial measurement system i.e. IMU incorporated to its design for the measurement of the orientation, acceleration and forces on the UAV in the three axes. The gimbal system can be used to test the UAV as a whole or just its control unit [20]. A major advantage of the gimbal test rig is its ability to allow free movement of the UAV in all the three-axes. The gimbal rig can also be used to test only the control unit making the rig valuable in the design phase. The limitations of the rig include the limitations of the sizes of UAVs to be tested to the size if the rings used for the gimbal. The rig is also susceptible to a phenomenon known as gimbal lock, that happens when all the three axes of the gimbal align.

Thrust stands (Figure 2.5 (a)) have also been used for the testing of UAVs. They provide the advantage of being able to test UAV parts i.e. the propellers and motors. However, this is also their major limitation as the stands cannot be used for whole UAVs.

(a)

(b)

Figure 2.5: Types of UAV test stands (a) Thrust stand [21] (b) Gimbal test rig [22]

### 2.2.3 Use of simulations

Simulations are research method technique that imitates an actual or theoretical system by means of an analogous mathematical model [23]. This is done with the aim of visualizing the behaviour of the part under test under a set of environmental conditions [15].

Computational Fluid Dynamics (CFD) is a branch of fluid mechanics that uses numerical analysis and data structures to solve problems that involve fluid flow. CFD software can be used together with other techniques such as the use of wind-tunnels for validation of data obtained from the method. There has been work done using CFD analysis for the analysis of the propellers of a quadcopter by Saleh and Abdelrady [24]. The results obtained especially for the wake structures of the UAV model were consistent with those obtained by experiment.

Finite Element Method (FEM) is most often used for structural analysis to show the effects of loads in a dynamic environment. Zhang et al [25]. used FEM analysis to do a drop analysis for small, light UAVs. The work done was on the comparison of simplified UAV models to both the highly detailed UAV models and experimental processes. The results were that even the simplified models gave results that were accurate enough for the acceptance of the simplified models.

The parameters tested in a UAV range from its performance characteristics i.e maximum speed, range of flight, maximum thrust and payload, electrical characteristics such as the power consumption, sensor reliability and battery levels to the communication protocols of the UAV. This, however, is not an exhaustive list and other tests depending on the intended function of the UAV can also be designed e.g. image quality test for UAVs used for photography [26]. The cost of testing UAVs, especially in closed controlled environments has been a challenge. This is because commercial test rigs are expensive with the cost of thrust stands being about Kes. 350,000 (LY-10KGF Thrust Stand) [27]. Wind tunnels are also quite costly. There is also a challenge of procuring test services locally with few recognised developers in the country [3] [28].

## 2.3 Test bed parameters

UAV test of choice depends on the parameter of interest. For example, a thrust stand is used to determine the aerodynamic properties of UAV propellers during the design phase

[29] and the navigation properties of a drone tested as a measure of the synergy of the UAV hardware and software elements [30].

### 2.3.1 Orientation

Orientation is the angular displacement of an object from a certain reference (axis) [31]. Orientation is important to a UAV since it determines the path to the UAV is going to take during flight. The sensors used to estimate the orientation of an object are the gyroscope sensors that measure the angular velocity of an object [32]. The result from the gyroscope is then mathematically integrated to obtain the orientation of the object. The orientation measured by gyroscopes, as shown on Figure 2.6, can be classified as yaw, pitch and roll. Yaw is the horizontal rotation of an object as seen from above, roll is the horizontal rotation of an object as seen from the front and pitch is the vertical rotation of an object as seen from the top. Orientation of a UAV can be represented as axis-angle

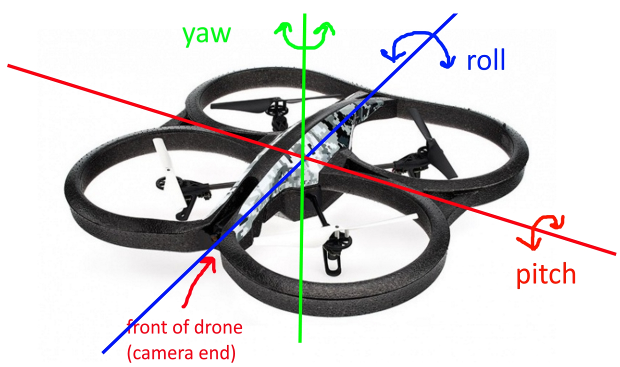



Figure 2.6: Orientation of a UAV [33]

representation, quaternions or Euler's angles representations.

Axis-angle Representation representation uses two parameters i.e a unit vector that defines the reference axis and the angle of rotation about the axis [34]. Thus in this system any orientation, as shown in Figure 2.7, can be represented by four parameters i.e. $\Theta, \hat{x}, \hat{y}, \hat{z}$ where $(\Theta)$ is the amount of rotation about the axis $(\hat{x}, \hat{y}, \hat{z})$. Axis-angle representations have a major limitation of being computationally unstable when x-axis and the y-axis align, also known as gimbal lock [36].

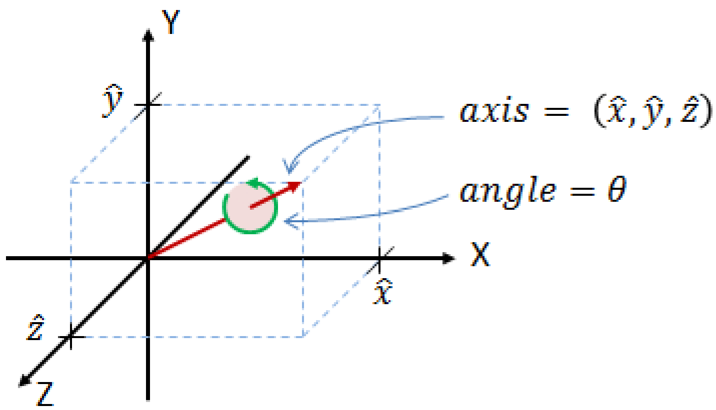



Figure 2.7: Axis-angle representation [35]

Quaternions use four components for orientation representation [35]. The components are composed of one real component and three imaginary components that represent the unit vectors shown on Equation 2.1.

$$q = q_0 + \mathbf{i}q_1 + \mathbf{j}q_2 + \mathbf{k}q_3 \tag{2.1}$$

The real part represents the magnitude of rotation and the other three represent the axis about which the rotation takes place. Quaternions are suitable for UAV testing when the calculating of rotation transformations is required as it is computationally faster than the other methods described. They are, however, more complex to quickly interpret [36].

Euler's angles define an orientation represented as a series of three rotations $\alpha$, $\beta$ and $\gamma$ about x, y and z axes [37] as shown on Figure 2.8. This representation is suitable for UAV testing to quickly tell the orientation of an UAV, since the three angles given by the representation give the roll, pitch and yaw of the UAV under test.

Electrical gyroscopes can either be analogue or digital in terms of their output and can be further classified in terms of the number of axes they can provide per measurement [38].

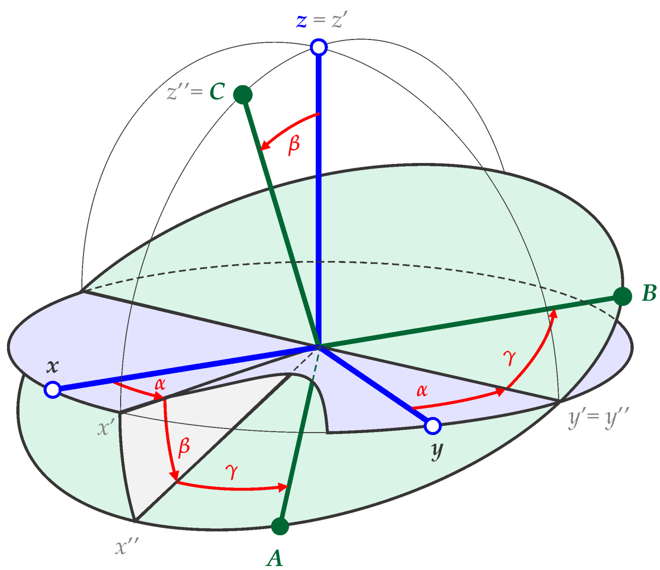



Figure 2.8: Euler's angles representation [37]

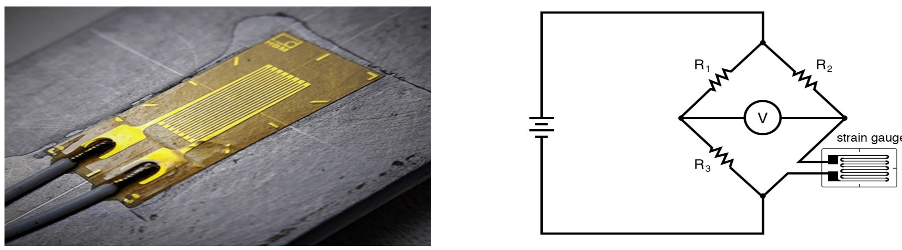
The gimbal rig features a gyroscope attached to the holding platform, in addition to the column stand, also having a gyroscope at the top of the column system. The gyroscopes in these systems are used to measure the angular velocity and the orientation of the UAV during the test.

### 2.3.2 Force

For a UAV to achieve flight, it must exert some thrust force. Forces of these nature as studied by Novotnak et al. [39] can be measured by the use of test rigs by the use of force sensitive resistors, strain gauges and load cells.

Force-sensitive resistors are measurement devices that convert the force applied to them to a change in their resistance.

Strain gauges, shown on Figure 2.10 (a), are used to measure the strain on objects and by extension the force on the object. Strain is the deformation on an object due to applied stress and, by extension, force. The gauges work by converting a tensile or compression force acting on a test material that results in a change in resistance of the gauge and this change in resistance is what is used to determine the amount of force acting on the material. This change is then read off from the sensor, and since the force applied is directly proportional to the strain on the object, the force on the object can be obtained [40]. A Wheatstone bridge, as shown on 2.10 (b), may be used to read from the strain gauge due to its sensitivity to low electrical signals. After reading the electrical signal, the signal may be further amplified and conditioned to remove noise from the signal.




(a) (b)

Figure 2.10: Strain Gauge (a) Strain Gauge Exterior [41] (b) Quarter Bridge Circuit [42]

Strain gauges have been used on UAV test rigs to test the thrust parameters of UAVs. This has been done by attaching the strain gauges to the structures holding the UAV during flight [39].

Load cells are measurement devices that convert mechanical force to another measurable output. The load cells may be classified based on the force they measure e.g. tensile and compressive force or the output they produce e.g. hydraulic, pneumatic and electric. Electrical load cells, most commonly work by the use multiple strain gauges bonded to a structure that deforms when force is applied. The force applied is then converted to electrical signals by the strain gauges and read off the instrument [43]. Load cells in test rigs are used in the measurement of the thrust parameters of UAVs on the test beds by using them as the attachment point between the UAV holding platform and the rest of the rig structure [44].

A comparison between the load cells and the strain gauges is shown in Table 2.1 [45].

Table 2.1: Load cells and strain-gauges

| | Load Cell | Strain Gauge |
|---|---|---|
| Measurement Range | Few grams of force to hundreds of tonnes | Up to a few thousand kilograms |
| Environmental Considerations | Robust and resistant to harsh environmental conditions | Sensitive to environmental conditions |
| Cost | Comparatively more expensive from their complex designs and higher accuracy | Are more cost-effective and thus suitable for budget-oriented applications |
| Maintenance and Lifespan | Longer lifespan with relatively little maintenance required | Requires occasional calibration and maintenance due to their volatility. |

### 2.3.3 Acceleration

Linear acceleration is the rate of change of the linear speed of an object. This parameter is measured by accelerometers, that produce electrical signals proportional to the acceleration on the sensor [46]. The accelerometer uses spring-mass system, which produces a force based on the acceleration or vibration experienced on the device. The force is converted to an electrical signal as the output of the sensor. Accelerometers in testing rigs are used to measure the acceleration of UAVs on the rigs. The acceleration can be used as a measure of the rate of ascent and descent of the UAV, in addition to being used as a quick overview of the responsiveness of the UAV under measurement [44]. Depending on the means of conversion, accelerometers may be divided into groups such as capacitive, Hall-effect and piezo-electric accelerometers [38]. Accelerometers may also be classified according to the number of axes covered simultaneously, giving rise to one-axis and three-axes accelerometers.

There has been work done by JKUAT students, Hunja and Nduati [44] on the making of a UAV testing rig . The testing rig made was a test bed, that was used to measure force due to the UAV by means of load cells, the orientation of the UAV using a gyroscope and the acceleration of the UAV using an accelerometer. The strain gauges were mounted along the support frame of the bed and the rest of the sensors at the holding area of the UAV. The results of the sensors were then posted to a simple interface made on the computer for visualisation. The bed was also able to measure the orientation of the UAV by means of accelerometers. The design, however, had the limitation of the load cells being used being prone to failure and were thus unreliable for the measurement of force during the test period. The supports for the UAVs were also relatively stiff hindering the movement of the UAV. This was due to the springs used to attach the UAV mounting plate being too stiff for the rig.

## 2.4 Gap analysis

Based on the literature review, the following gaps in the UAV testing area were seen:

1. There are inadequate locally made reliable UAV testing rigs in the market forcing institutions and other clients to source the required equipment from outside the country.

2. The cost of acquiring UAV testing rigs is relatively high. These costs range in the hundreds to thousands of dollars, which is unaffordable for most institutions or individuals. There is therefore need for a locally fabricated and reliable UAV testing rig at an affordable price.

# 3 Methodology

The UAV testing rig has been designed to measure the orientation, thrust and acceleration of a UAV fastened onto the system. We chose a closed test mode of measurement in order to avoid any environmental interference such as air drag interfering with the system during measurement.

Applying a modular approach to the design, the mechanical module focuses on the physical structure for the UAV testing rig, the electrical module will tackle the power circuitry required by the rig while the control module will interpret and present the values to the users.

For this project, the general procedure for the use of the testing rig is as follows:

1. The UAV being tested is loaded onto the testing plate and fastened.
2. Sensors are initialized and zeroed to compensate for the added weight of the UAV on the system.
3. The UAV is turned on and oriented in a desired direction as it manipulates the plate to move along with it and the sensors measure their configured parameters (strain gauges for force, the gyroscope for orientation and the accelerometer for acceleration).
4. The data from the sensors is collected and displayed on the user interface.
5. The UAV may be unloaded if the test is complete or another run be performed if any changes are made.

## 3.1 Mechanical module

The following are the general considerations made in the design of the mechanical structure of this project:

1. A geometric shape that would evenly distribute forces from the UAV holder for accurate measurements
2. A hydraulic diameter of 1000 mm to accomodate different UAV sizes for testing. The UAV sizes range from diagonal wingspans of 378 mm (e.g a DJI Mini 2 SE [2]) to 1172 mm (e.g an EFT G06 V2 UAV Agricultural Spraying Drone [47])
3. Rigid supports around the frame to withstand the forces involved in addition to minimization of clutter in the design
4. Low weight for portability

The mechanical module design is thus divided into the following main parts:

1. Test Rig Frame
2. Test Rig Support
3. UAV Plate
4. Springs
5. Material Considerations

### 3.1.1 Test Rig Frame

Its geometry was driven by the number of strain gauges that would be utilized in the system and the need to obtain accurate readings without additional calculations i.e. all eight strain gauges should give the same readings for a uniformly distributed load. Circular and octagonal frames were evaluated as shown on Table 3.1.

Table 3.1: Shape design considerations

| Design requirement | Circular frame | Octagonal frame |
|---|---|---|
| Mounting | This mostly relies on cylindrical rods that require bending in order to achieve the circular geometry. The circular geometry does not provide suitable mounting surfaces for the plates, due to the plates only having tangential points of contact with the plates | Can be achieved by the use of angle lines cut at specific dimensions to achieve the required hydraulic diameter. This will also provide a flat surface for the mounting of the aluminium plates |
| Stress points | They have induced stressed points that may lead to mechanical failure with repeated use | Stress is distributed more evenly across adjoining metal pieces |
| Cost | Bending a metal sheet/rod into a circular shape is costlier | Relatively less expensive in comparison as all that would be required is cutting and welding |

Looking at the considerations on Table 3.1, an octagonal frame was chosen for the project as shown on Figure 3.1.

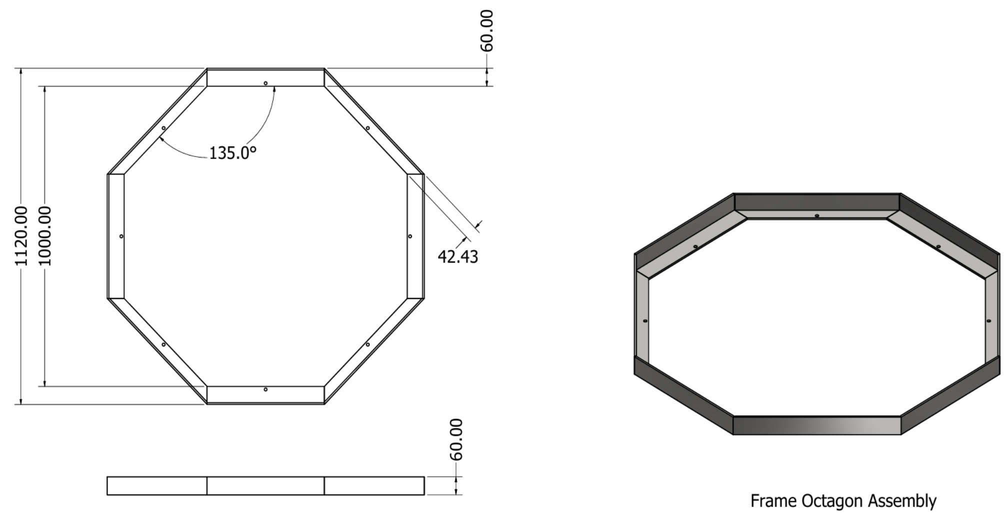



Figure 3.1: Octagonal Frame

The joining technique that was chosen for the frame is arc welding. This is because of its widespread use and availability as well as its ability to make permanent rigid joints.

Three materials, Aluminium, Mild steel and Stainless steel, were considered for the octagonal frame. The factors considered for material selection of the Test Rig frame are shown on Table 3.2.

Table 3.2: Frame material considerations

| Property | Mild steel | Aluminium | Stainless steel |
|---|---|---|---|
| Strength | High Impact strength | Low impact strength | Excellent impact strength |
| Density (g/cm$^3$) | 7.85 | 2.7 | 8 |
| Cost | Inexpensive | Moderately expensive | Expensive |

By comparing the three, Stainless steel outweighs the mechanical properties of Mild Steel with stainless steel also having a shiny finish thus also aesthetically pleasing. Stainless steel is however more expensive than aluminium or mild steel. Aluminium, although lighter than stainless steel and mild steel, is also weaker than the two metals as it is more susceptible to physical alterations from a moderate force applied.

Mild steel was thus chosen for the frame as it was both strong enough while remaining affordable. The octagonal frame would be made up of angle lines with $60 \times 60 \times 6$ mm for weight reduction, and ease of attachment with the rest of the system

### 3.1.2 Test Rig support

Since the frame of the testing rig is octagonal in shape, angle lines with 135° will be required to interface with the edges of the frame. Four supports were chosen to reduce the clutter on the design, as shown on Figure 3.2, while maintaining strength and balance on the rig. A height of 300 mm for the supports was chosen to ensure the suspended UAV does not touch the ground during testing.

The supports will be mounted on a square metal sheet with dimensions of $150 \times 150 \times 5$ mm to improve the stability and the rigidity of the test rig. This is shown on Figure 3.3. The four support assemblies will be joined to the frame using arc welding.

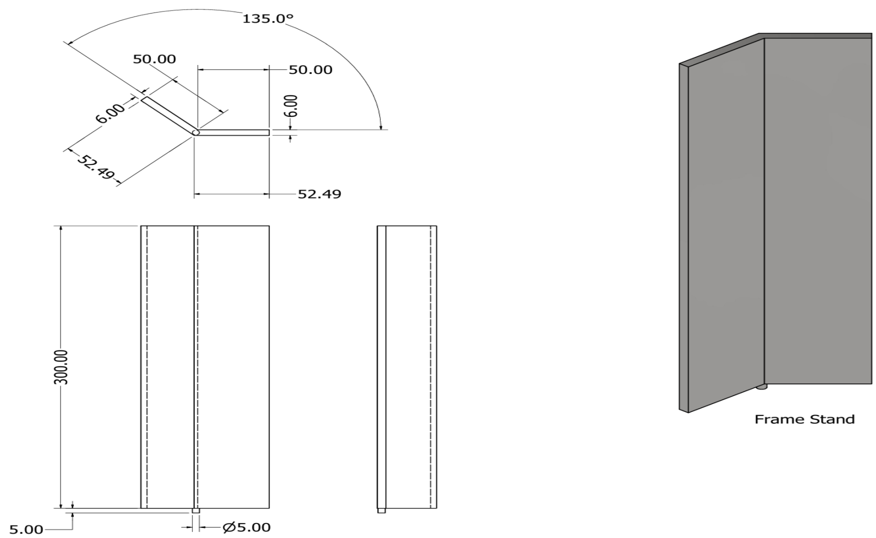



Figure 3.2: Support Frame

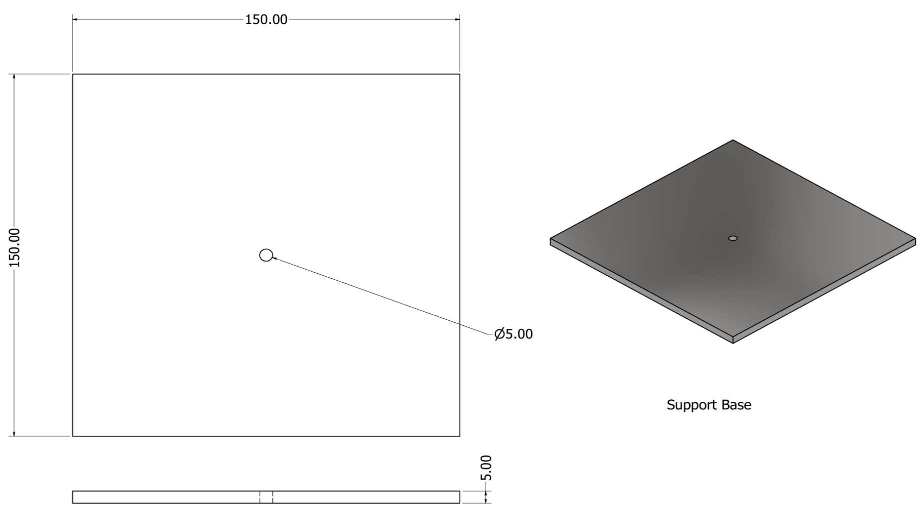



Figure 3.3: Support Base

### 3.1.3 Sensor Plate

Strain gauges require a surface to be mounted on for accurate measurement that is also flexible enough to have some strain imparted on it. Therefore, an aluminium strip would suit for this application as it is malleable as compared to mild steel. It has dimensions of $15 \times 8 \times 2$ mm making it thin enough to strain under load. It also has two 10 mm holes in diameter, one which will be used to fasten the strip to the frame by screws and the other to hook an extension spring that goes to the plate of the mechanism.

### 3.1.4 UAV plate

The plate will be used to hold the UAV during the test. Its design was based on the number of measuring points to be utilized by the system, the shape of the frame, its weight to reduce its effect on the UAV movement and the reduction of the downwash exerted by the UAV propellers during the test. This resulted in a star-shaped plate, shown on Figure 3.4, with eight points that have holes of 20 mm in diameter each to be attached with extension springs to the sensor holders making it suspended in air like a trampoline. The plate has an overall dimension of 700 mm hydraulic diameter and a thickness of 9 mm. The middle section has a rectangular hole of $240 \times 200$ mm for the UAV mounting mechanism with four 10 mm diameter holes for mounting the holding mechanism as shown on Figure 3.4. The material considerations made for the part are

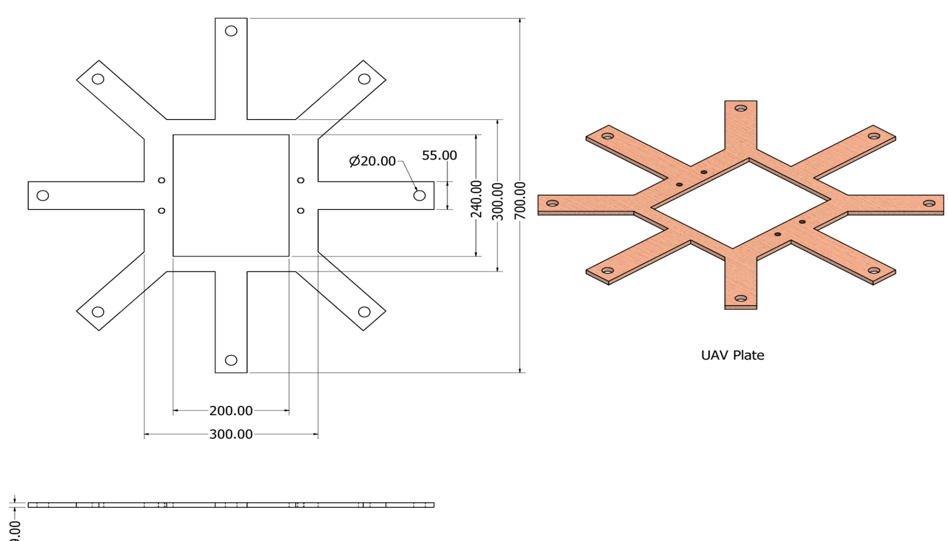



Figure 3.4: UAV Plate

summarised on Table 3.4.

Table 3.3: UAV Plate Material considerations

| Property | Aluminium | Plywood |
|---|---|---|
| Strength (MPa) | 32 | 13.8 |
| Density ($g/cm^3$) | 2.7 | 0.63 |

Plywood was thus chosen for the UAV plate due to its sufficient strength with an expected maximum force of 60N applied on the test bed for the UAV tests to be conducted, while still being lighter than aluminium.

### 3.1.5 UAV Holding Mechanism

This comprises of three slider designs coupled with a sliding support that will be used to hold the UAV in place during parameter measurement fitting at the centre square hole of the plate. Two sliders have an overall dimension of $192 \times 25 \times 9$ mm with threaded through holes of 8 mm such that the tightening of these sliders will be done by stud screws. One of these two has the hole at the centre (centre slider) while the other has its hole near one of its ends (end slider). The third slider has an added lip of $5 \times 2$ mm on one long edge such that when paired with a centre hole slider, the lip will act as a base support for the UAV so that its landing gear does not protrude below leading to incorrect measurement as the UAV slides up and down (lip slider). The correct arrangement of sliders for mounting from each end will be as follows: centre slider to lip slider to end slider from each end. The centre slider also has 10 mm in diameter threaded through holes that will be used to fasten it in place with the sliding support. The supports are shown on Figure 3.5

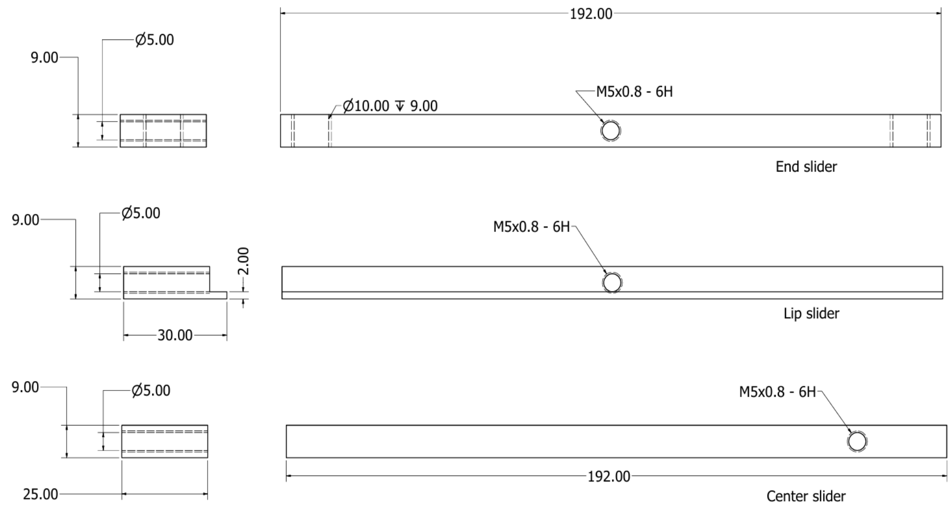



Figure 3.5: Sliding supports

The slider holder, shown on Figure 3.6 will hold the sliders to the plate using screws to keep the UAV in place. It has dimensions of 200 × 52 × 13 mm with a cross-sectional design based off a cantilever rack. It has four through holes that will be used to keep the centre sliders in place and mount them to the plate.

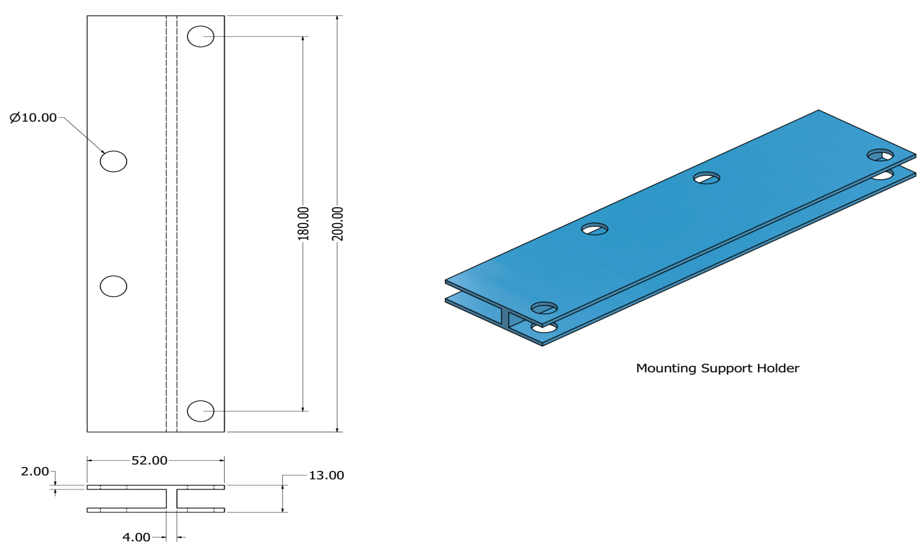



Figure 3.6: Slider Holder

Table 3.4: Mounting Support Material Considerations

| Property | PLA Plastic | Plywood |
|---|---|---|
| Strength (MPa) | 26.082 | 13.8 |
| Density ($g/cm^3$) | 1.25 | 0.63 |

PLA plastic was chosen for the part due to its low density, affordability and the ability to 3-D print it. 3-D printing was chosen for production of the part due to the complexity of the holding mechanism in addition to the relatively small size of the same.

### 3.1.6 UAV Holding Mechanism simulation

Finite Element Analysis, as shown on Figure 3.7, was performed on the UAV plate assembly i.e. the UAV plate and holding mechanism to test the strength of the structure in addition to determining the suitability of the materials chosen for the sliding mechanism (PLA plastic) and UAV plate (plywood) while working with a total load of 60 N separated into two 30 N point loads placed on the point of attachment of the UAV landing gears to the holding mechanism.

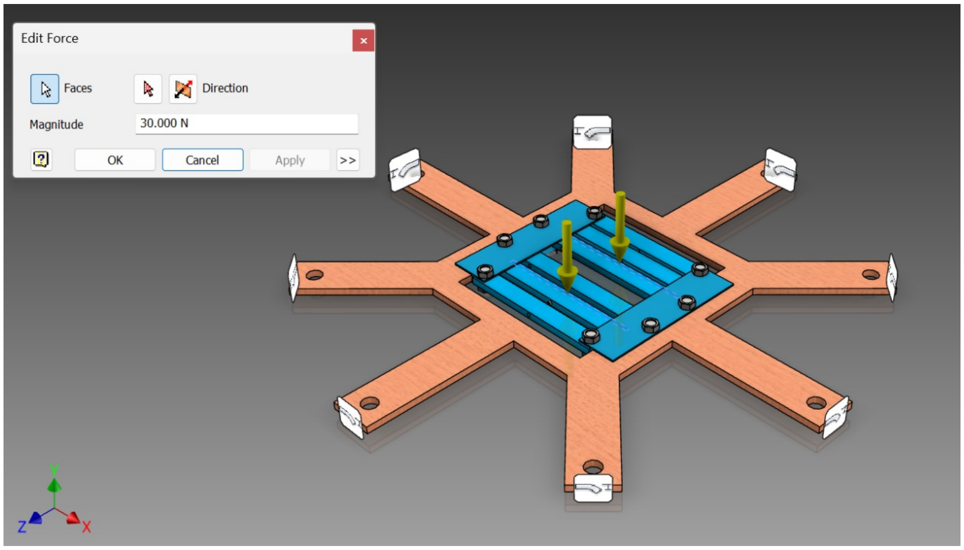



Figure 3.7: Simulation setup

### 3.1.7 Springs

Springs were incorporated into the design of the test rig to allow free movement of the UAV on its platform during testing and to transmit force from the platform to the strain gauges for the measurement of force. The following were the main considerations in the design of the spring:

1. The spring should be a maximum of 100 mm long to fit in the space between the holder platform and outer structure of the rig with minimal extension.
2. The spring should have a maximum extension of about 5 mm, when the UAV is at its maximum thrust, to prevent too much play on the holder during the test
3. The attachments at the end of the springs should be easily attachable to the holder platform during assembly.
4. The spring should be locally available.

With an expected extension of 5 mm, and an expected load of about 29.43 N (13.84 N for the holder platform and 15.59 N for maximum weight of the UAV), k was calculated using Equation 3.1.

$$F = k\Delta x \tag{3.1}$$

Where:

$F$ = force on one spring. The force is applied equally on eight springs, thus for one spring the force is equal to 3.67875 N

$k$ = spring constant

$x$ = extension of 0.05 m

The spring constant is obtained as 73.575 N/m

The wire diameter and mean diameter were calculated using Equation 3.2.

$$k = \frac{G\ d^4}{8D^3 N_a} \tag{3.2}$$

Where:

$G$ – shear modulus ($79 \times 10^9 GPa$)

$d$ – Wire diameter

$D$ – mean diameter

$N_a$ – Number of active coils

$k$ – Spring constant

$$73.575 = \frac{79\ \times\ 10^9\ \times\ d^4}{8\ \times\ D^3\ \times\ N_a}$$

After substituting the known values to Equation 3.2, Equation 3.3 was obtained.

$$\frac{73.575\ \times\ 8\ \times N_a}{79\ \times\ 10^9} = \frac{d^4}{D^3} \tag{3.3}$$

Equation 3.4 was used to relate the number of active coils to the wire diameter and length of the spring.

$$\frac{L_o}{d} - d = N_a \tag{3.4}$$

Where:

$L_o$ = length of spring obtained as 100 mm from the design requirements

$d$ = Wire diameter

$N_a$ = Number of active coils

The wire diameter was determined by by linearly varying the wire diameter, in steps of 0.05 mm from 0 - 0.4 mm, and obtaining both the c, required to be a value between 5-10 for ease of manufacture, and the number of active coils for ease of handling in Equation 3.2 and 3.3 as shown on Equation 2.4. 0.04 mm was thus seen as a compromise between the two factors i.e. mean index of 5.988 and number of active coils of 250. Calculating D (mean diameter), using c (spring index) and d (wire diameter) gives a mean diameter of 2.4 mm

The spring parameters are thus shown on Table 3.5

Table 3.5: Spring parameters

| Parameter | Value |
|---|---|
| Mean diameter | 2.4 mm |
| Wire diameter | 0.4 mm |
| Spring Constant | 73.575 N/m |
| Material | Mild steel |
| Unloaded length | 100 mm |

### 3.1.8 Sensor Plate

A sensor plate, Figure 3.8 was designed for attaching the UAV platform to the surrounding structure and act as a measuring point for the strain gauges used in the rig.

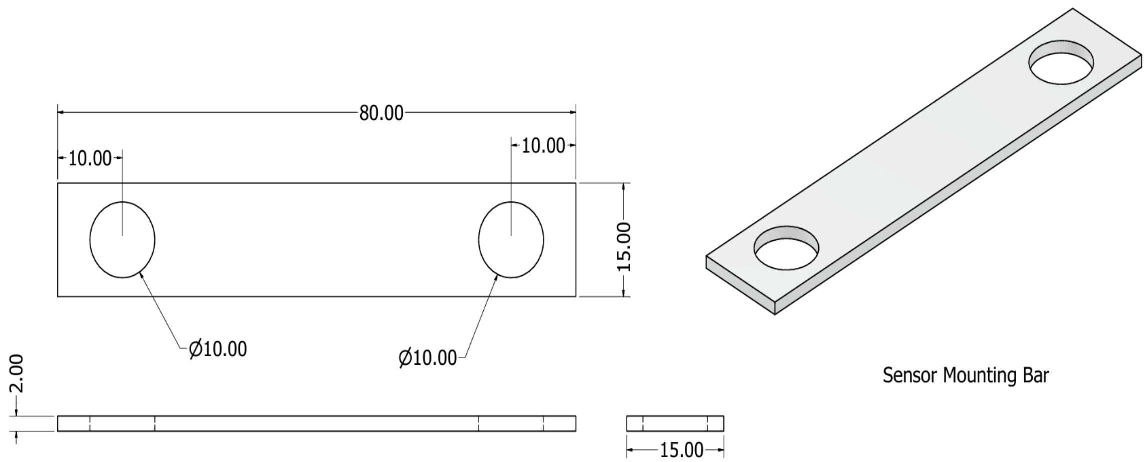



Figure 3.8: Sensor plate

### 3.1.9 Material consideration for Mechanical Design

A summary of the properties of the materials considered is shown in Table 3.6.

Table 3.6: Material comparison summary

| Part | Material | Density ($g/cm^3$) | Yield Stength (MPa) | Advantages |
|---|---|---|---|---|
| Frame, Main support | Mild Steel | 7.85 | 250-650 | Affordable, High Strength |
| Plate | Plywood (Birch) | 0.68 | 13.8 | Lightweight, Rigid for the intended purpose |
| Sensor bar | Aluminium | 2.7 | 32 | Flexible, Lightweight |
| Mounting Supports | Plastic (PLA) | 1.24 | 26 | Easy to print, Sufficient strength for the intended purpose |

## 3.2 Electrical module

The electrical module comprises of the electrical components that will make up the testing rig and the considerations that went into choosing the electrical system. The power requirements of each component chosen and the means by which the components will be powered is considered.

### 3.2.1 Power supply requirements

For the power supply, the main consideration were as follows:

1. The power supply should be able to supply all the components reliably, giving a voltage of 5V.
2. The power supply chosen, should be built into the system.
3. The power supply, should provide a constant current and voltage. This is required to supply the strain gauge with constant power for accurate measurements.

The overall power requirements for the system are shown on Table 3.7

Table 3.7: Power requirements for electrical components

| Component | Number | Power requirement |
|---|---|---|
| ESP32 Micro-controller | 1 | 3.3-3.6 V, 500 mA |
| HX-711 amplifier | 8 | 5 V,1.6 mA each |
| Strain Gauge | 8 | 5 V, 1.6 mA each |
| MPU-6050 gyroscope | 1 | 3.3-5 V 3.9 mA |
| LM2596 Buck Converter | 1 | 3-40 V input, 1.5-35 V output at 2 A |
| LM044L LCD screen with I2C adapter | 1 | 2.4-5 V 100 $\mu$A |
| Total | | 5 V, 518.7 mA |

The total power consumption was to 2.6 W. Two 9V rechargeable dry cells were chosen as the main power source. They were chosen due to their current rating of 300 mAh, that would provide enough current for the device for about 2 hours before needing a recharge [48]. The batteries are also portable and can be re-charged making them economical. A battery was considered to cater for portability and measurement in areas without electricity. An adjustable buck converter was also incorporated into the circuit to step down the voltage to 5V, acting as a voltage regulator for the circuit.

### 3.2.2 Strain gauge configuration

The main consideration for the use of strain gauge was the ability to accurately measure the change in resistance. The strain gauge will be connected in a Wheatstone bridge configuration in the quarter bridge configuration, as shown in Figure 3.9. The quarter-bridge circuit was chosen mainly because of its relatively easy implementation while providing accurate measurements. A variable resistor will be used on one side of the bridge to balance the bridge during fabrication.

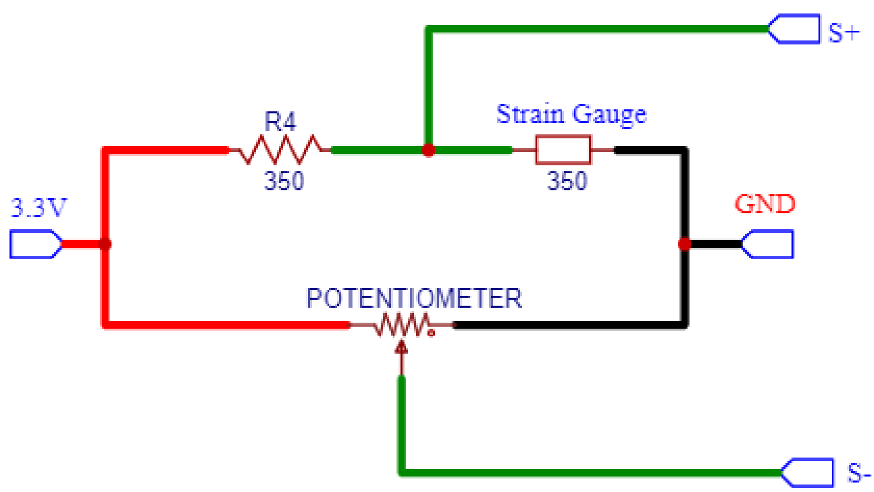



Figure 3.9: Strain Gauge circuit

The provision of a constant current and the ability to read the output of the bridge were also considered for the design. The solution to both these challenges was the use of the HX711 A/D converter to both power and recieve input from the strain gauges used.

## 3.3 Control module

The control module involves the logic for running the testing rig in addition collecting and processing of data from the rig. The user interface and its various functions is also included in the module.

### 3.3.1 Measurement methods

Acceleration, force and orientation of the UAV are the parameters the rig will measure. Linear acceleration is the measure of the change of speed per unit time. The factors considered for the choice of the accelerometer for use are:

1. A sensor with power requirements that can be supplied by a microcontroller i.e 3.3 - 5 V and 0 - 400 mA.

2. A sensor that measures the acceleration in the three-axes for the movement of the UAV.

3. A sensor that is relatively easy to interface with common micro-controllers i.e. a digital output or an analog value of between 0 - 5volts.

4. A sensor that is relatively small and can easily be embedded onto the UAV holder.

5. The accelerometer should preferably have a gyroscope attached to the chip.

Based on these factors the sensors in Table 3.8 were considered [49][50][51]:

Table 3.8: Design considerations for accelerometers and gyroscopes

| | MPU6050 | ADXL335 | ADXL1001 |
|---|---|---|---|
| Power requirements | 2.4 - 3.5 V 3.9 mA | 1.8 - 3.6 V 350 $\mu$A | 3.0 - 5.25 V 1.15 mA |
| Measurement Axes | 3 axes | 3 axes | 1 axis |
| Output | I2C format | Analog 0 - 3.3 V | Analog 0 - 2.5 V |
| Dimensions | 21 mm $\times$ 16 mm $\times$ 8.76 mm | 19 mm $\times$ 19 mm $\times$ 3.14 mm | 5 mm $\times$ 5 mm $\times$ 1.8 mm |
| Gyroscope presence | Yes | Yes | Yes |

The MPU6050 was chosen due to its small size, the number of axes it can measure simultaneously and its digital output. The chip has an added advantage of having a gyroscope attached to it

Orientation is the angular positioning of the UAV with respect to the reference frame of the rig. This parameter will be measured by the use of a gyroscope that measures angular velocity. The angular velocity will then be mathematically integrated to obtain the orientation of the UAV in all the three axes. The representation chosen for this parameter is the Euler's angles representation, due to its intuitive nature and the little computation required in the test [37].

The force to be measured by the rig is the weight and thrust of the UAV. The thrust of the UAV includes its distribution between the four motors. The following considerations were made in the measurement of force in the rig:

1. High sensitivity of the measurement
2. Ability to measure both compressive and tensile forces on the UAV
3. Relative ease of integration of the measurement devices with the micro-controller
4. The cost of the instrument

Table 3.9 shows the instruments considered for force measurement on the rig [45].

Table 3.9: Comparison of force measuring instruments

| | Strain gauge | Load-cell | Force-dependent resistor |
|---|---|---|---|
| Power requirements | 3.3 - 5 V , for the amplifier | 3.3 - 5 V, for the amplifier | about 5 V |
| Type of force measured | Tensile and compressive force | Tensile and compressive force | Compressive force |
| Range of measurement | Upto a few thousand kilograms | Upto hundreds of tonnes | upto 100N (FSR 402) |
| Cost of implemetation | Kes. 300 | Kes. 3500 | Kes. 300 |

The strain gauge was chosen due to its affordability over the strain gauge and its ability to measure both compressive and tensile forces.

### 3.3.2 Communication protocol

Wireless communication would be required between the UAV rig, the micro-controller and the user interface for ease of use of the rig during normal operation. Serial communication would be implemented for debugging the rig and for diagnostic purposes.

The wireless technologies considered for the rig were bluetooth and wireless fidelity (Wi-Fi). Table 3.10 shows the comparison between the two technologies [52].

Table 3.10: Comparisons of wireless communication technologies

| | Wi-Fi | Bluetooth |
|---|---|---|
| Frequency Bands | 2.4 GHz and 5 GHz | 2.4 GHz (from 2402 MHz to 2480 MHz for BLE) |
| Data rates | Max speeds of about 9.6 Gbps | 2 Mbps |
| Power consumption | Low energy required, especially for BLE | High energy consumption |

Wi-Fi was chosen for the communication between the rig. In addition to this Wi-Fi technology was deemed as easier to implement due to the ease of use of its libraries, its integration with the user interface to be made and wireless communication between the system and the dtabase.

### 3.3.3 Data processing and display

This module involves the data handling once it has been acquired from the sensors, data storage for further processing and its display during the test.

The user interface was divided into two:

1. A 20 column by 4 row LCD interface that would be attached to the rig for off-line support of the machine. This is mainly to aid in the initialization phase, to display the general status of the machine and to aid in the correction of any errors should they occur.

2. A web-page, hosted on the user's computer, will display the measurements from the sensors. The user will be able to send commands for the zeroing of the sensors before the test can begin, the test start and the test end.

Data from the test-rig would be stored in a time-series database, since time is important in the analysis of the UAV parameters. The database would implement a comma-separated value (.csv) file format as opposed to other formats like JavaScript Object notation (.json). This is because the csv format is relatively more efficient to store large amounts of data with little hierarchical structure to them, which is our expected data format. The CSV format can also be imported into many computer applications, for example Matlab and Excel, for further analysis of the recorded data.

### 3.3.4 Micro-controller selection

A micro-controller is a programmable electric component used for the implementation of some logic in a circuit. The micro-controller for the rig would be responsible for the reading of all the sensor input to the system, the control of the LCD and communication between the user interface and the rig.

The design considerations for the micro-controller selection were:

1. A relatively high processing speed due to the tasks of the micro-controller in the rig
2. A large number of I/O pins to accomdate the large number of sensors in the design
3. Built-in wireless communication for the micro-controller
4. Affordability of the micro-controller
5. Presence of developer support for the micro-controller

The microcontrollers considered for the rig were the Arduino Mega , ESP32 and the Raspberry Pico. Table 3.11 shows the differences between the three microcontrollers [54][55][56].

Table 3.11: Micro-controller considerations

| | Arduino-Mega | Raspberry Pico | ESP-32 |
|---|---|---|---|
| Clock speed | 16 MHz | 133 MHz | 80-240 MHz |
| General I/O pins | 54 | 36 | 26 |
| Wireless communication | No built-in support | No built-in support | Supports both Bluetooth and Wi-Fi |
| Cost | Kes. 2000 | Kes 1200 | Kes 1000 |
| Developer support | Well supported | Limited support | Limited support |

Based on the comparisons on Table 3.10, the ESP32 board was selected due to its relatively large number of I/O pins, wireless communication capability and affordability.

# 4 Results

## 4.1 Mechanical module

The overall assembly is as shown in Figure 4.2 depicting the final look of the rig after the final assembly.

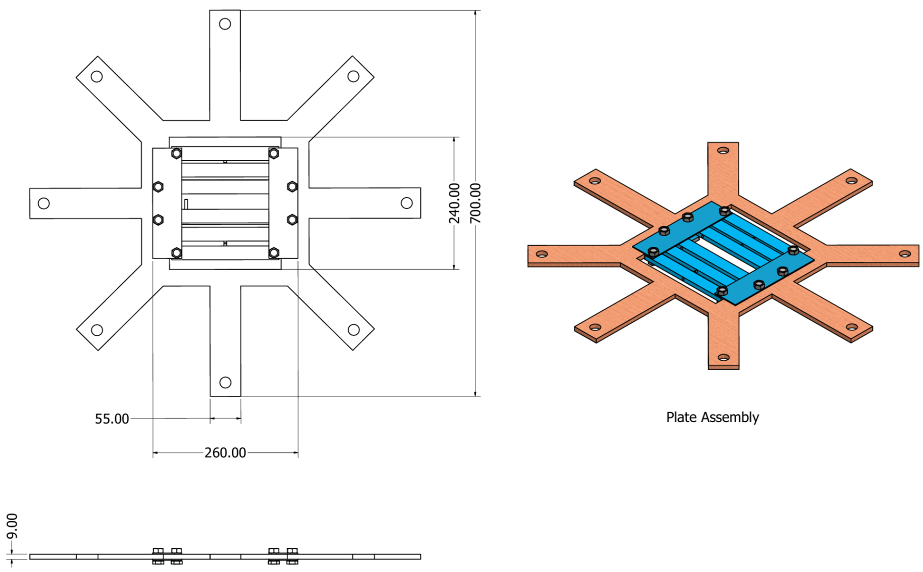



Figure 4.1: UAV test rig assembly model

This octagon frame was selected to provide equal distance from the center to each edge with a hydraulic length of 1000 mm to accomodate UAVs of different sizes. Mild steel was selected to provide both strength and rigidity to the frame. The total mass of the rig was 29 kgs for portability reasons. The assembly, shown on Figure 4.2 and Figure 4.3, had the UAV plate assembly and sensor plate as its main parts.

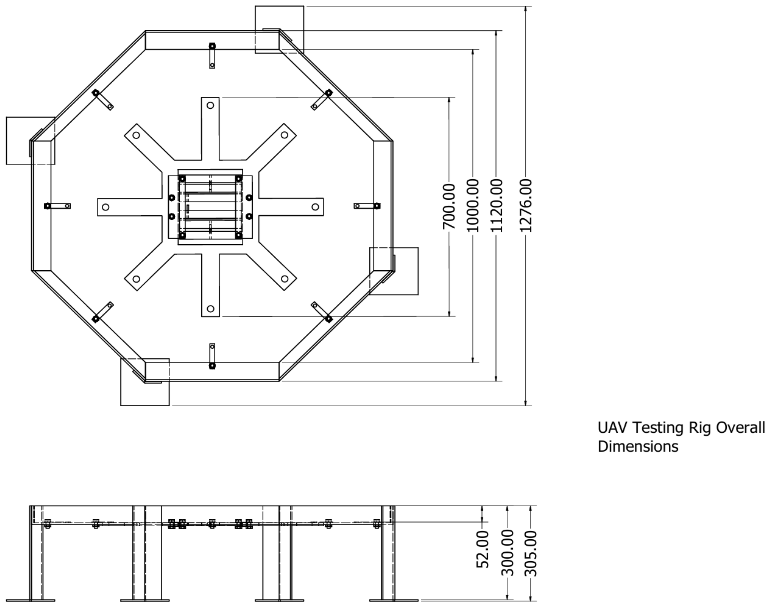
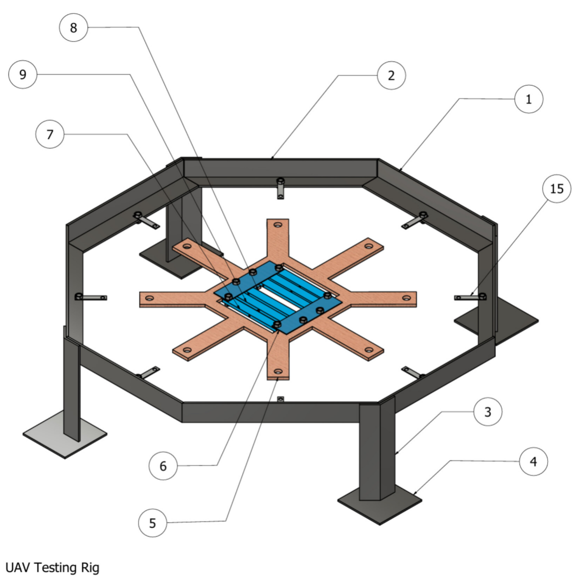



Figure 4.2: UAV test rig assembly




| PARTS LIST | | | |
|---|---|---|---|
| ITEM | QTY | PART NUMBER | MATERIAL |
| 1 | 4 | Base Apex | Steel, Mild |
| 2 | 4 | Base Diagonal | Steel, Mild |
| 3 | 4 | Frame Stand | Steel, Mild |
| 4 | 4 | Frame Stand base | Steel, Mild |
| 5 | 1 | UAV Plate | Plywood |
| 6 | 2 | UAV Mounting Holder | PLA Plastic |
| 7 | 2 | Sliding Bar 1 | PLA Plastic |
| 8 | 2 | Sliding Bar 2 | PLA Plastic |
| 9 | 2 | Sliding Bar 3 | PLA Plastic |
| 10 | 16 | AS 1427 - M10 x 20 | Steel, Mild |
| 11 | 16 | AS 1237 - 10 | Steel, Mild |
| 12 | 16 | AS 1474 - M10 | Steel, Mild |
| 13 | 2 | DIN 938 - M5 x 22 | Steel, Mild |
| 14 | 1 | DIN 835 - M5 x 18 | Steel, Mild |
| 15 | 8 | Sensor holder | Aluminum 6061 |

Figure 4.3: UAV test rig assembly model with parts list

**UAV Plate Assembly**

This structure, shown on Figure 4.1 is used to mainly used to constrain the UAV to the rig during the test process, while allowing free movement of the UAV. It is made up of the UAV plate, shown in 4.1, and the mounting assembly for the UAV.

## 4.2 Stress Analysis Results

Simulations were done on the UAV mounting plate. This was done by applying two point loads of 30 N on the mounting points for the UAV. The results of this are shown on Figure: 4.4.

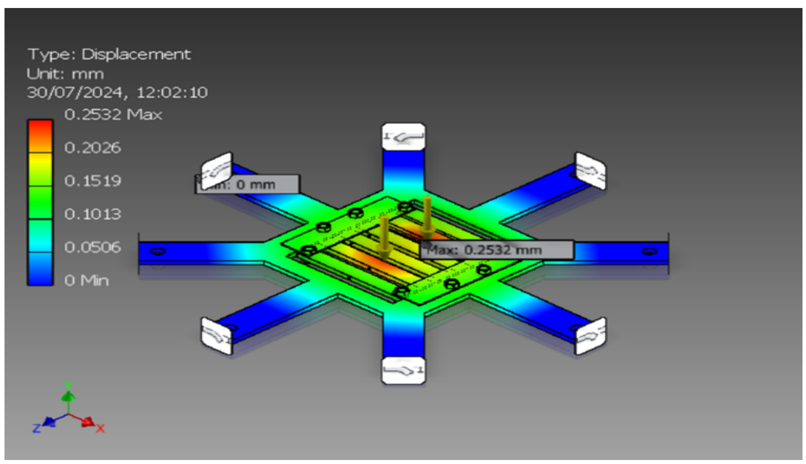



Figure 4.4: UAV plate simulation results

The stress obtained from the simulation was relatively equally distributed throughout the holder with its maximum value being 83.3385 MPa. A maximum deflection of 0.253 mm was also obtained. The materials, plywood, for the UAV plate, and PLA plastic, for the mounting assembly, were thus deemed sufficient for this application. The combined mass of the UAV plate and mounting supports was also an important design consideration. This mass came to 0.825 kg.

The detailed technical drawings for the rig, including the remaining sub-assemblies and individual parts, have been attached to Appendix B.

## 4.3 Electrical module

Given the design considerations and choice of electrical components, an electrical circuit was designed for use in the rig.

The rig will have a total power consumption of about 2.6 W, with two - 9 V batteries used to provide power for the rig. The major sub assemblies for the circuit are the strain gauge circuit and powering circuit.

### 4.3.1 Strain gauge circuit

The main considerations for the strain gauge connections was the ability to provide a constant current to the gauges and the amplification of the signals from the gauges. Based on the above considerations the circuit on Figure 4.5 was designed.

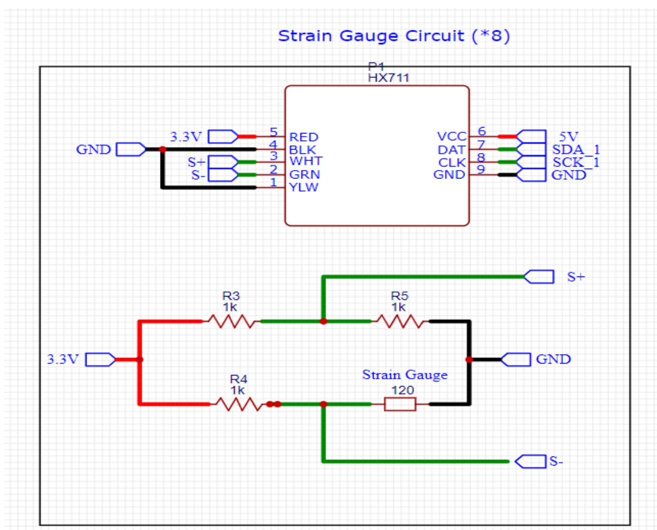



Figure 4.5: Strain gauge circuit

### 4.3.2 Powering circuit

The ability to provide reliable power to all the electrical components in the testing rig was the main consideration in the design of this circuit. The circuit, shown on Figure 4.6 was thus designed. A switch was also incorporated to the design for turning the rig on and off.

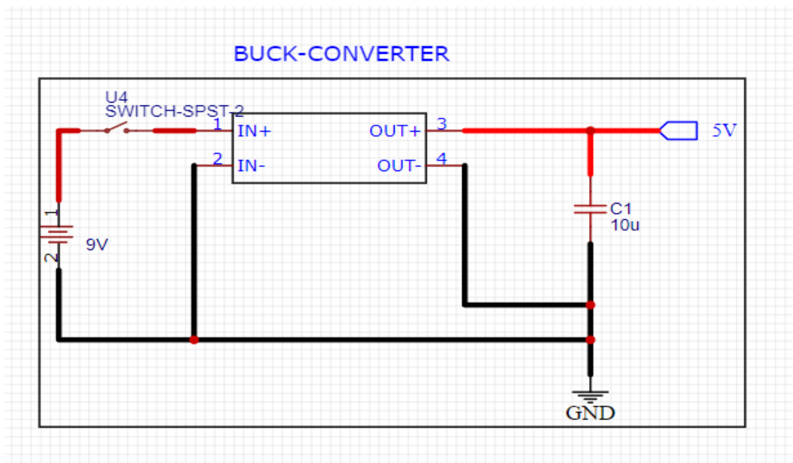



Figure 4.6: Powering circuit

The detailed circuit has been attached to Appendix D.

## 4.4 Control module

The control module consists of the logic of the testing rig, the programme implementation and the user interface for interaction with the UAV rig.

The control module design yielded the logic flowchart shown in Figures 4.7 and 4.8 in addition to a preliminary user interface.

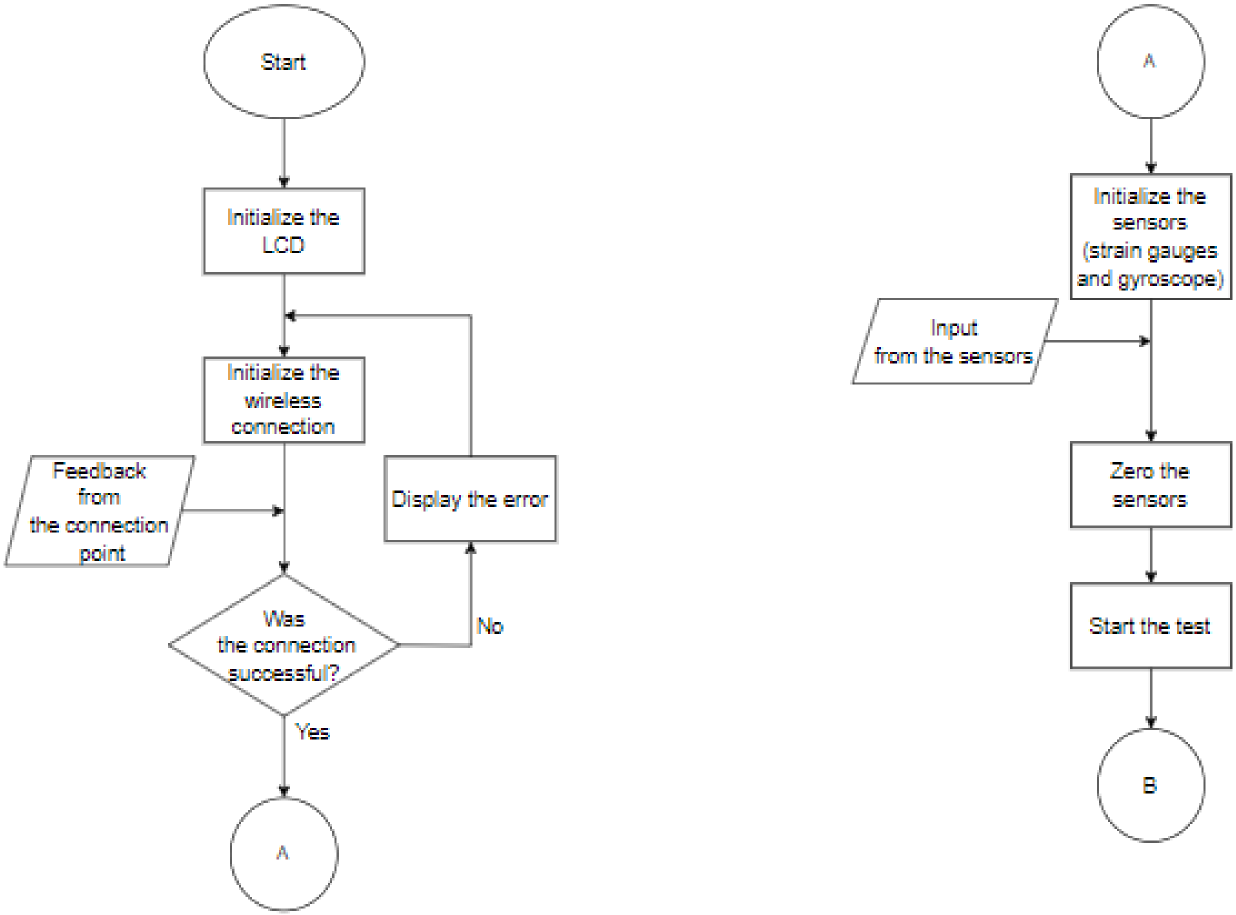



Figure 4.7: Control flowchart (Initialization)

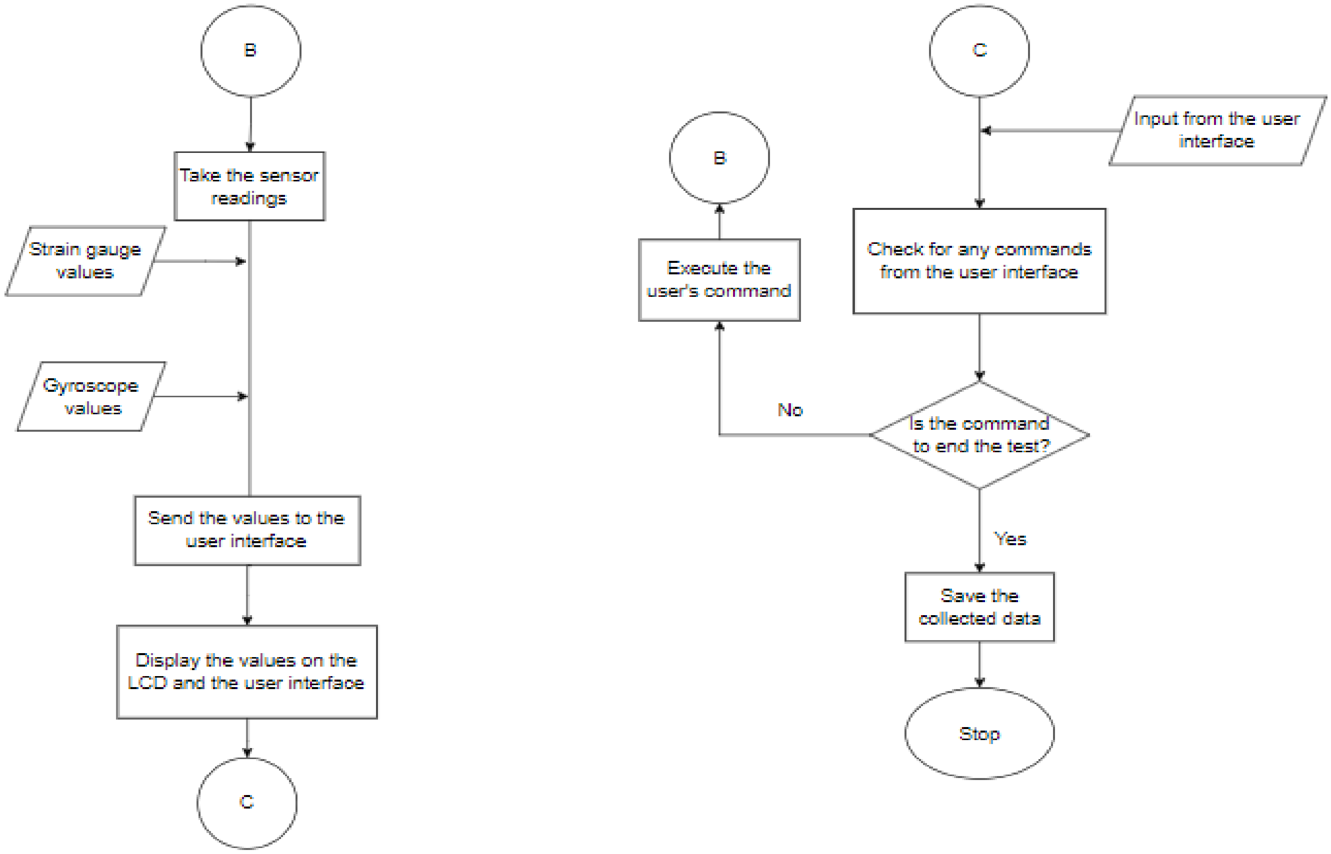



Figure 4.8: Control flowchart (Testing)

## 4.5 Preliminary Budget

Table 4.1: Preliminary budget

| Number | Item | Description | Quantity | Unit Price | Total (Kes) |
|---|---|---|---|---|---|
| 1 | Angle-line | 2” by 2” by 0.25 (3 m) | | | 3550 |
| 2 | Aluminium-6061 | Sheet | 0.25 | 350 per kg | 88 |
| 3 | PLA Roll | 1.75 mm (1 kg) | 1 kg | 3500 per kg | 3500 |
| 4 | Springs (mild steel) | 100 mm unloaded length | 8 | | 500 |
| 5 | Strain-Gauge | BF-350-3AA | 8 | 150 | 1200 |
| 6 | Amplifier + A/D Converter | HX-711 amplifier | 8 | 200 | 1600 |
| 7 | Accelerometer + Gyroscope | MPU-6050 | 1 | 400 | 400 |
| 8 | Micro-controller | ESP32 | 1 | 1500 | 1500 |
| 9 | LCD | LM044 (2004) | 1 | 700 | 700 |
| 10 | Buck Converter | LM2596 | 1 | 500 | 500 |
| 11 | Batteries (Dry-cells) | 9 V (2-Pack Energiser) | 2 | 500 | 1000 |
| | TOTAL | | | | 14,538 |

# 5 Conclusion

The mechanical, electrical and control design of the test rig has been presented. This is in addition to the design considerations and component selection process. The mechanical design for the rig comprises an octagon-shaped frame that uses angle lines in its build and supports, with an eight-point star-shaped plate for mounting the UAV onto. The electrical design comprises the electronic components used to power the rig and achieve functionality based on requirements, with components selected such as the types of sensors to be implemented and the power delivery system. The control design comprises the micro-controller choice that will enable logic control over the rig as well as the system flow chart illustrating how the rig will operate.

# References

[1] Consortiq "A Not-So-Short History of Unmanned Aerial Vehicles (UAV)," Consortiq, Available: https://consortiq.com/uas-resources/short-history-unmanned-aerial-vehicles-uavs [Accessed June. 20,2024].

[2] DJI Official Website Available: https://www.dji.com/global/mobile/mini-2-se-specs [Accessed August. 20,2024].

[3] KenDrone "Commercial Drone Services" KenDrone Available: -https://www.kendrone.co.ke/ [Accessed August. 20,2024].

[4] Dynotis "The Importance of Test Stands for Drone Efficiency Testing" Dynotis, Available: https:// dynotis.semai.com.tr/the-importance-of-test-stands-for-drone-efficiency-testing [Accessed June. 30,2024].

[5] Drone Academy-Asia "What is the difference between a Drone, a UAV and a UAS," Drone Academy-Asia, Available: https://www.droneacademy-asia.com/post/what-is-the difference-between-a-drone-a-uav-and-a-uas [Accessed June. 20, 2023].

[6] Built In "Types of Drones" Built In, Available: www.builtin.com/articles/types-of-drones [Accessed Jun. 20,2024].

[7] ScienceDirect "Fixed Wing Drone," ScienceDirect, Available:https://www.sciencedirect.com/topics/computer-science/fixed-wing-drone [Accessed August. 20,2024].

[8] AirSight "Types of Drones - Fixed Wing," AirSight, Available: https://www.airsight.com/learn/airspace-security/drone-fundamentals/types-of-drones-fixed-wing [Accessed June. 20,2024].

[9] Unmanned Systems Technology "Unmanned Helicopters and Helicopter Drones," Unmanned Systems Technology, Available:

https://www.unmannedsystemstechnology.com/company/alpha-unmanned-systems/alpha-800-uav-helicopter/ [Accessed August. 20,2024].

[10] D-shop.com ”4K HD Wide-angle Quadcopter Remote Control Drone with Camera,” D-shop.com , Available: https://www.dshop.com.au/buy/4k-hd-wide-angle-quadcopter-remote-control-drone-w/E88PRO-B [Accessed June. 20,2024].

[11] Umiles ”Types of drones. Classification by use and characteristics” Umiles, Available: https://umilesgroup.com/en/types-of-drones-classification-by-use-and-characteristics [Accessed August. 20,2024]

[12] Dekra ”Drone and Unmanned Aerial Vehicle Testing and Certification,” Dekra, Available: https://www.dekra.com/en/drones-and-unmanned-aerial-vehicle-testing-and-certification/ [Accessed June. 20,2024].

[13] M Vierhauser K Meixner S Biffl ”Scenario-Based Field Testing of Drone Missions” Euromicro Conference on Software Engineering and Advanced Applications (SEEA) 2024

[14] Pawel Cwiakala ”Assessment of accuracy of basic maneuvers performed by an unmanned aerial vehicle during autonomus flight,” Faculty of Mining Surveying and Environmental Engineering, AGH University of Science and Technology, August. 2017.

[15] Kate M. Pavlock ”AeroSpace Engineering Handbook: Chapter 2(v): Flight Test Engineering” National Aeronautics and Space Administration, Dryden Flight Research Center

[16] J. Foster L. Miller R. Busan S. Langston ”Recent NASA Wind Tunnel Free-Flight Testing of A Multirotor Unmanned Aircraft System Vehicle” NASA Langley Research Center, Hampton

[17] V. Pandey "Wind Tunnel: A Review," International Journal of Innovative Research in Science Engineering and Technology, Department of Mechanical Engineering, Vivekanada Global University, Jaipur India, February. 2017.

[18] Medium "How crucial is a drone test stand in UAV development," Medium, Available: https://medium.com/@shimanshiji22/how-crucial-is-a-drone-test-stand-in-uav-development-26c759325888/ [Accessed June. 30,2024].

[19] Choi Hyo-seon, "Testing bed for drone," South Korea Patent KR101853623B1, June. 08,2018.

[20] VectorNav "What is an Inertial Measurement Unit," VectorNav, Available: https://www.vectornav.com/resources/inertial-navigation-articles/what-is-an-inertial-measurement-unit-imu [Accessed June. 20,2024].

[21] Klein Educational Systems "What is an Inertial Measurement Unit," Klein Educational Systems, Available: https://www.kleineducational.com/blog/102-minds-i-drone-gimbal-rig-perfect-for-testing-indoors [Accessed June. 20,2024].

[22] Wing Flying Tech "LY-30KGF Thrust Stand," Wing Flying Tech, Available: https://www.wingflyingtech.com/motor-test/ly-30kgf-thrust-stand [Accessed June. 20,2024].

[23] IGI Global "What is Scientific Simulation" Available: https://www.igi-global.com/dictionary/scientific-simulation

[24] A. Saleh, E. Abelrady "CFD analysis of full quadcopter" International Journal of Advanced Multidisciplinary Research and Studies, Department of Mechatronics, Faculty of Engineering, November 2022.

[25] Y. Zhang, Y. Huang, Z. Li, K. Cao "A Simplified FE Modelling Strategy for the Drop Process Simulation Analysis of Light and Small Drone" Aerospace 2021, 8, 387. Available "https://doi.org/10.3390/aerospace8120387 [Accessed August. 20,2024]

[26] Biokinetics "What Factors are assessed with drone testing,"Biokinetics, Available:https://biokinetics.com/what-factors-are-assessed-with-drone-testing [Accessed August. 20,2024].

[27] FoxTech "LY-10KGF Thrust Stand" FoxTech Available:https://www.foxtechfpv.com/ly-10kgf-propulsion-system-test-platform.html [Accessed August 20. 2024]

[28] DroneSpace " Home Page" DroneSpace Available: https://www.dronespace.co.ke/ [Accessed August.21 2024]

[29] P. Kosa, M. Kisev, L. Vacho, L. Toth, M. Olejar, M. Harnicariva, J. Valicek, H. Tozan "Experimental Measurement of a UAV Propeller's Thrust" Technical Gazette 29, 1(2022) Available https://doi.org/10.17559/TV-20201212185220 [Accessed August 21. 2024]

[30] Andrzej Tomczyk "In-flight tests of navigation and control system of unmanned aerial vehicle" Aircraft Engineering and Aerospace Technology December 2003

[31] DBpedia "Orientation (geometry)," DBpedia, Available: https://www.dbpedia.org/page/Orientation(geometry) [Accessed June. 25,2024].

[32] Elprocus "Gyroscope Sensor Working and its Applications" Elprocus, Available: https://www.elprocus.com/gyroscope-sensor [Accessed June. 25,2024].

[33] The Droning Company "Drone Flying for Dummies," The Droning Company, Available: https://www.thedroningcompany.com/blog/drone-flying-for-dummies

[34] CueMath "Rotation Matrix," CueMath, Available: https://www.cuemath.com/algebra/rotation-matrix/ [Accessed June. 25,2024].

[35] DancewithCode "Rotation Quaternions , and How to Use Them," DancewithCode, Available: https://danceswithcode.net/engineeringnotes/quaternions/quaternions.html [Accessed June. 25,2024].

[36] Hughes Noel, " What are the advantages and disadvantages of representing orientation in different ways" ResearchGate Available:https://www.researchgate.net/post/What-are-the-advantages-and-disadvantages-of-representing-orientations-in-different-ways/63b7ffdfe62569b205098a55/citation/download [Accessed August. 20,2024]

[37] EasySpin "Rotation and Euler angles" EasySpin Available:https://easyspin.org/documentation/eulerangles.html [Accessed August. 21,2024]

[38] Mouser Electronics "Gyroscope Types, Selection" Mouser Electronics Available: https://eu.mouser.com/applications/gyroscope-types-selection/ [Accessed August. 21,2024]

[39] J. Novotnak M. Firko P. Lipovsky and M.Smelko "Design of the system for Measuring UAV Parameters" Drones6, no 8: 213 Available: https://doi.org/10.3390/drones6080213 [Accessed August. 21,2024]

[40] Michigan Scientific Corporation "What is a Strain Gauge and How Does it Work," Michigan Scientific Corporation, Available: https://www.michsci.com/what-is-a-strain-gauge/ [Accessed June. 25,2024].

[41] Medium "Strain Gauge: Principle, Types, Features and Applications" Medium, Available: https://encardio.medium.com/strain-gauge-principle-types-features-and-applications-357f6fed86af [Accessed June. 26,2024]

[42] All About Circuits "Strain Gauges," All About Circuits, Available: https://www.allaboutcircuits.com/textbook/direct-current/chpt-9/strain-gauges/

[43] Omega "Load Cells: Introduction to Load Cells" Omega, Available: https://www.sea.omega.com/tw/prodinfo/loadcells.html [Accessed June. 25,2024].

[44] F. Hunja and L. Nduati,"Design and Fabrication of a Drone Test Bed," Jomo Kenyatta University of Agriculture and Technology,December 2022.

[45] Measurex ”Load Cells vs. Strain Gauges: Exploring the Differences and Benefits,” Measurex, Available: https://www.measurex.com.au/load-cells-vs-strain-gauges [Accessed June. 20,2024]

[46] Bruel and Kjaer Sound and Vibration Movement ”Accelerometer sensors” Bruel and Kjaer Sound and Vibration Movement, Available: https://www.bksv.com/en/transducers/vibration/accelerometers [Accessed June. 25,2024].

[47] RC-Wing ”EFT G06 V2 EFT G06 V2 4 AXIS 6L UAV Agricultural Spraying Drone Farm Sprayer,” RC-Wing Available: EFT G06 V2 4 AXIS 6L UAV Agricultural Spraying Drone Farm Sprayer [Accessed August. 20,2024]

[48] Nerokas ”Energizer Recharge 9V Rechargeable Battery” Nerokas, Available: https://store.nerokas.co.ke/SKU-1373 [Accessed August.10 ,2024]

[49] Last Minute Engineers ”Interface MPU6050 Accelerometer and Gyroscope Sensor with Arduino” Last Minute Engineers, Available: https://lastminuteengineers.com/mpu6050-accel-gyro-arduino-tutorial/ [Accessed August.10,2024]

[50] Last Minute Engineers ”How Accelerometer works? Interface ADXL335 Accelerometer with Arduino” Last Minute Engineers, Available: https://lastminuteengineers.com/adxl335-accelerometer-arduino-tutorial/ [Accessed August.10,2024]

[51] Seeeds Studio ”ADXL1001/ADXL1002 Datasheet” Seeeds Studio, Available: https://www.seeedstudio.com/Grove-Single-Axis-Analog-Accelerometer-100g-ADXL1001-p-4035.html [Accessed August. 10,2024]

[52] Moko Smart ”Bluetooth Vs WiFi Vs ZigBee: Which Wireless Technology is better” Moko Smart Available: https://www.mokosmart.com/bluetooth-vs-wifi-vs-zigbee-which-is-better [Accessed August. 10,2024]

[53] RcBeacon ”Microcontroller Power Supplies,” RcBeacon, Available: https://www.rcbeacon.com/pmb/psu_general_1 [Accessed June. 26,2024]

[54] Arduino Store ”Arduino Mega 2560 Rev3” Arduino Store Available: https://store.arduino.cc/products/arduino-mega-2560 [Accessed August. 20,2024]

[55] rfsolutions ”ESP32 WIFI MODULE” rfsolutions Available: https://www.rfsolutions.co.uk/downloads/6253ea295dd0e843DS-ESP32-1.pdf [Accessed August. 20,2024]

[56] M. Claussen, ”Pico Power.Get to know the Raspberry Pi Pico board and RP2040” rfsolutions Available: https://www.rfsolutions.co.uk/downloads/6253ea295dd0e843DS-ESP32-1.pdf [Accessed August. 20,2024]

# A Preliminary Budget

Table A.1: Preliminary budget

| Number | Item | Description | Quantity | Unit Price | Total (Kes) |
|---|---|---|---|---|---|
| 1 | Angle-line | 2" by 2" by 0.25 (3 m) | | | 3550 |
| 2 | Aluminium-6061 | Sheet | 0.25 | 350 per kg | 88 |
| 3 | PLA Roll | 1.75 mm (1 kg) | 1 kg | 3500 per kg | 3500 |
| 4 | Springs (mild steel) | 100 mm unloaded length | 8 | | 500 |
| 5 | Strain-Gauge | BF-350-3AA | 8 | 150 | 1200 |
| 6 | Amplifier + A/D Converter | HX-711 amplifier | 8 | 200 | 1600 |
| 7 | Accelerometer + Gyroscope | MPU-6050 | 1 | 400 | 400 |
| 8 | Micro-controller | ESP32 | 1 | 1500 | 1500 |
| 9 | LCD | LM044 (2004) | 1 | 700 | 700 |
| 10 | Buck Converter | LM2596 | 1 | 500 | 500 |
| 11 | Batteries (Dry-cells) | 9 V (2-Pack Energiser) | 2 | 500 | 1000 |
| | TOTAL | | | | 14,538 |

# B Time Plan

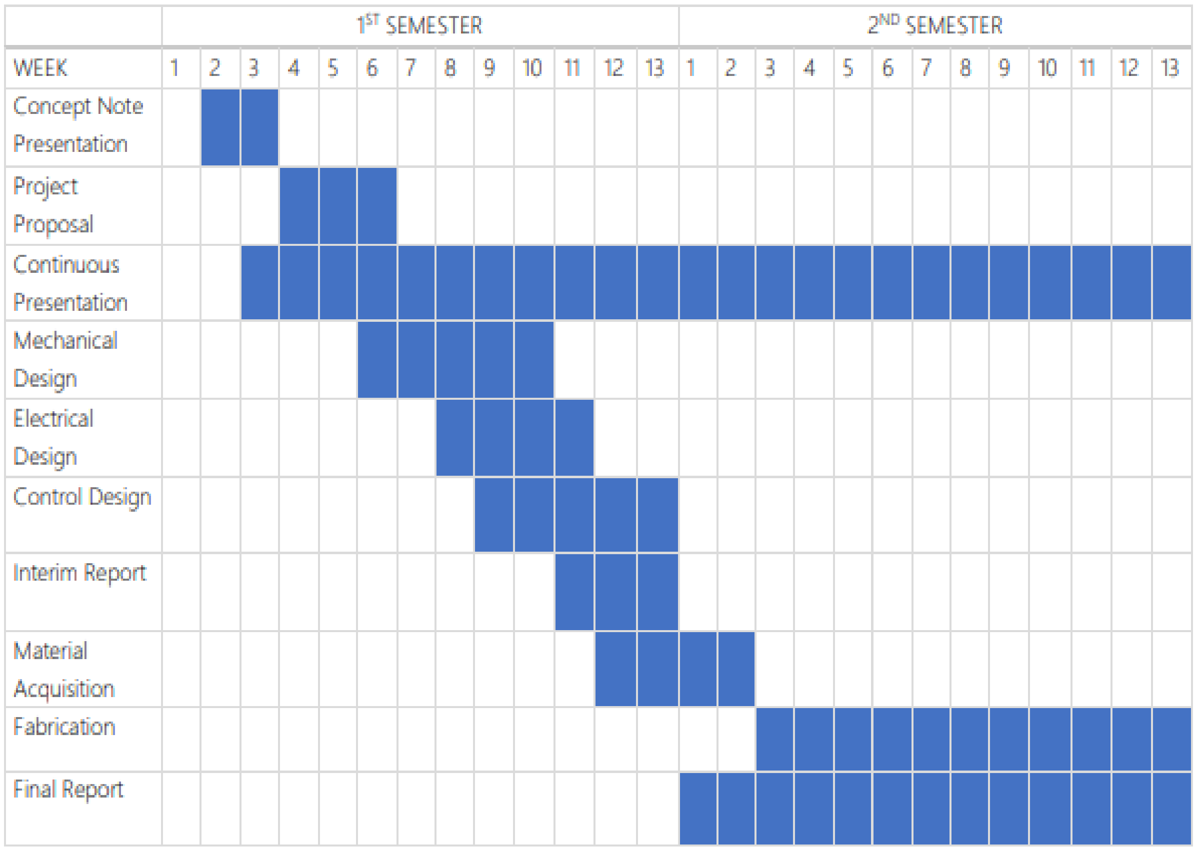



Figure B.1: Timeplan

# C Mechanical module

## C.1 Final assembly

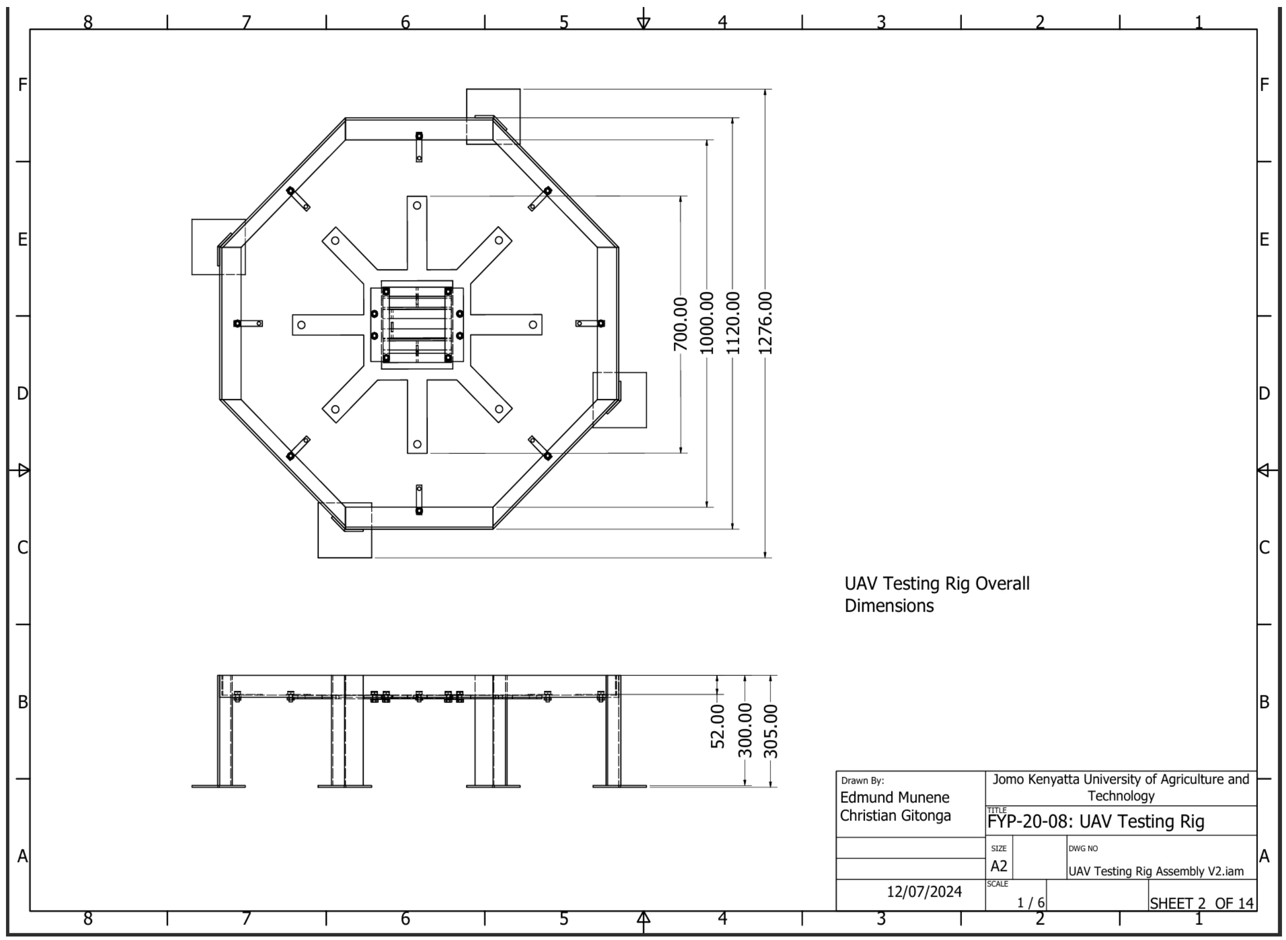



Figure C.1: Detailed test rig assembly

| PARTS LIST | | | | |
|---|---|---|---|---|
| ITEM | QTY | PART NUMBER | DESCRIPTION | MATERIAL |
| 1 | 4 | Base Apex | Cardinal Angle Lines for the octagon frame | Steel, Mild |
| 2 | 4 | Base Diagonal | Sub-cardinal Angle Lines for the octagon frame | Steel, Mild |
| 3 | 4 | Frame Stand | Main support for the rig | Steel, Mild |
| 4 | 4 | Frame Stand base | Added stability for the stand | Steel, Mild |
| 5 | 1 | UAV Plate | Where the UAV will be placed | Plywood |
| 6 | 2 | UAV Mounting Holder | Holds the sliding bars to the plate | PLA Plastic |
| 7 | 2 | Sliding Bar 1 | For keeping the UAV in place | PLA Plastic |
| 8 | 2 | Sliding Bar 2 | For keeping the UAV in place | PLA Plastic |
| 9 | 2 | Sliding Bar 3 | For keeping the UAV in place | PLA Plastic |
| 10 | 16 | AS 1427 - M10 x 20 | ISO metric machine screws | Steel, Mild |
| 11 | 16 | AS 1237 - 10 | Flat metal washers for general engineering purposes (metric series) | Steel, Mild |
| 12 | 16 | AS 1474 - M10 | Hex Nut | Steel, Mild |
| 13 | 2 | DIN 938 - M5 x 22 | Studs with metal end of 1d | Steel, Mild |
| 14 | 1 | DIN 835 - M5 x 18 | Double End Stud | Steel, Mild |
| 15 | 8 | Sensor holder | | Aluminum 6061 |

Figure C.2: Parts List

## C.2 Octagon frame assembly

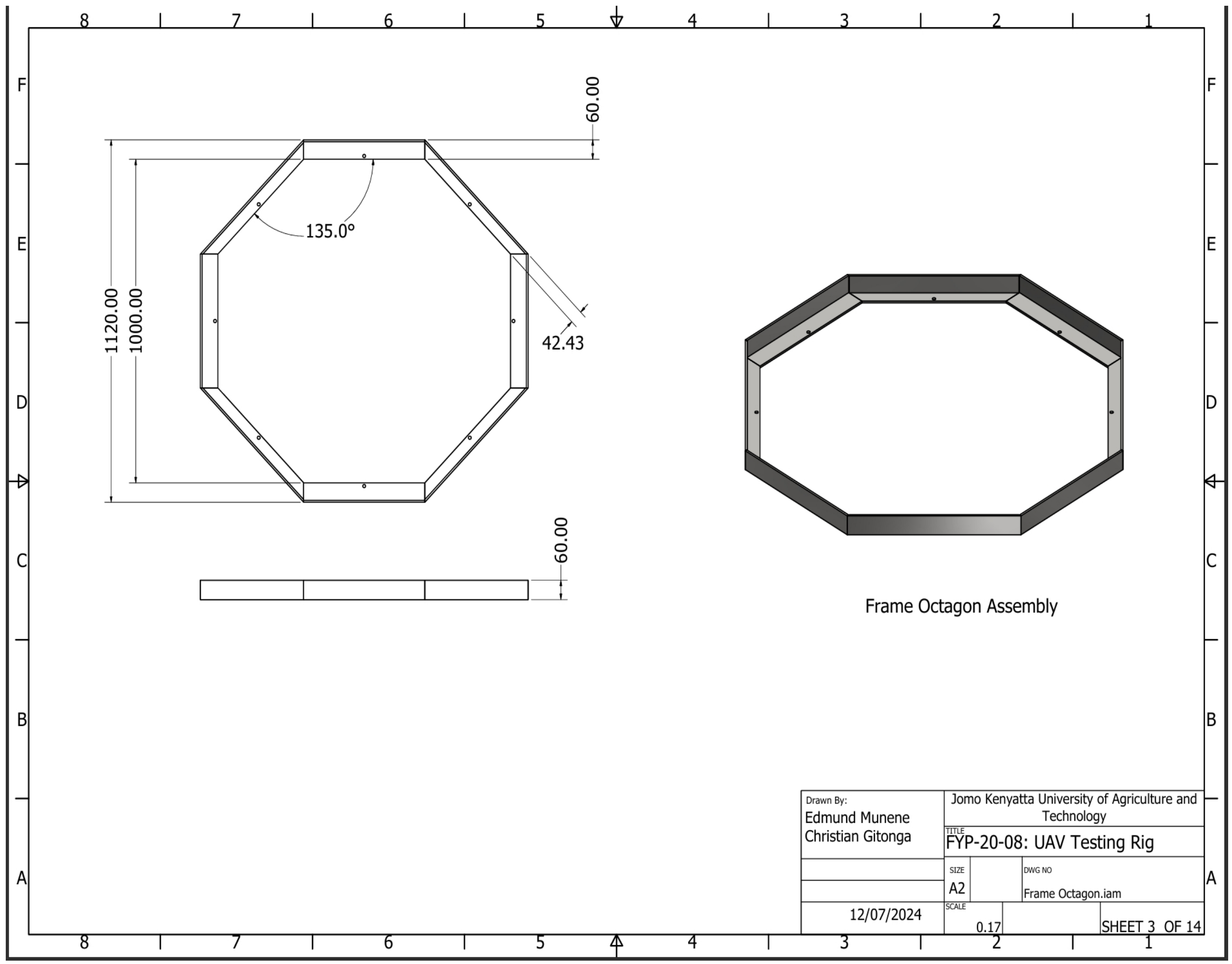



Figure C.3: Octagon frame assembly

### C.2.1 Frame Apex

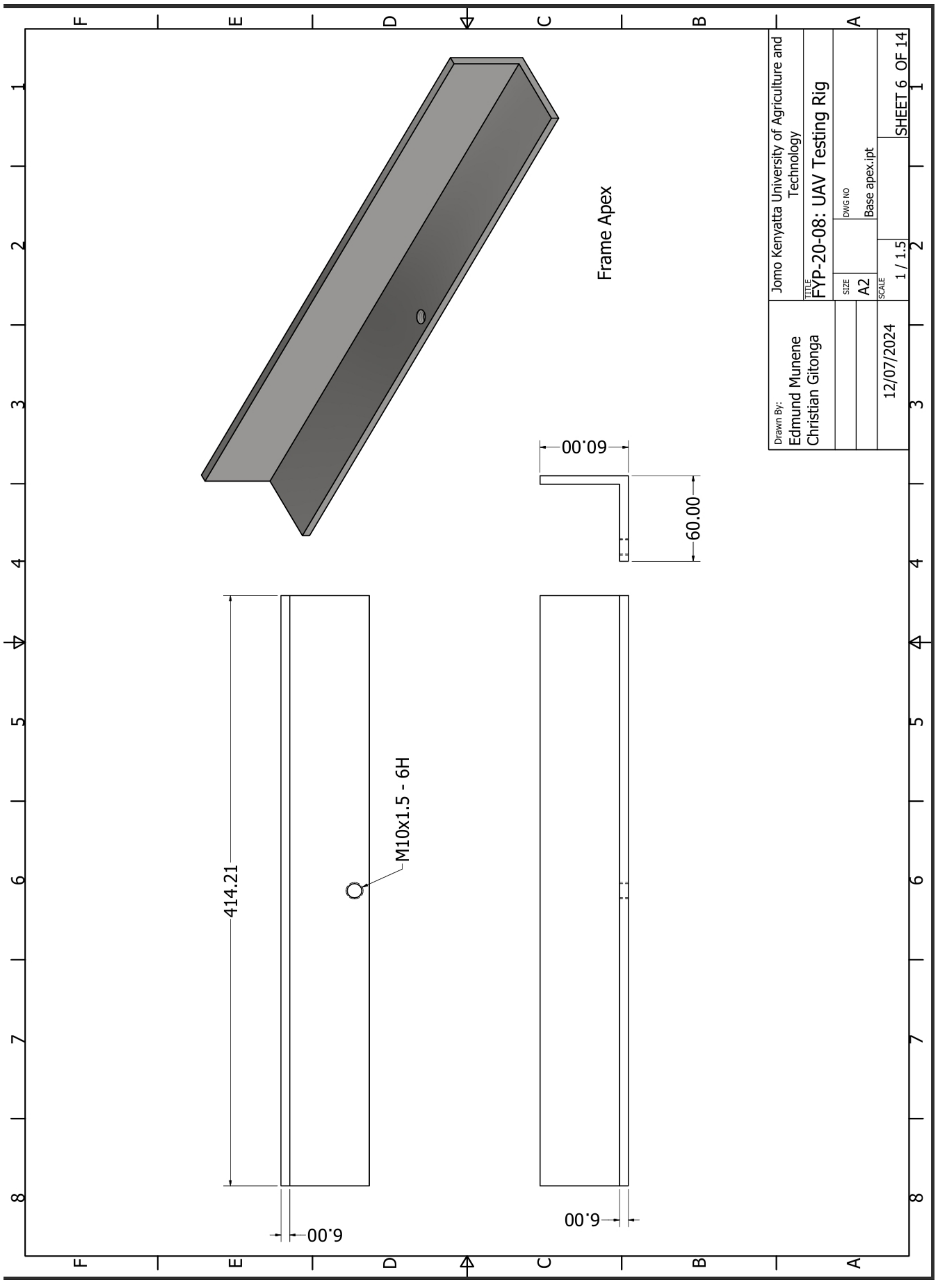



Figure C.4: Frame apex

### C.2.2 Frame Diagonal

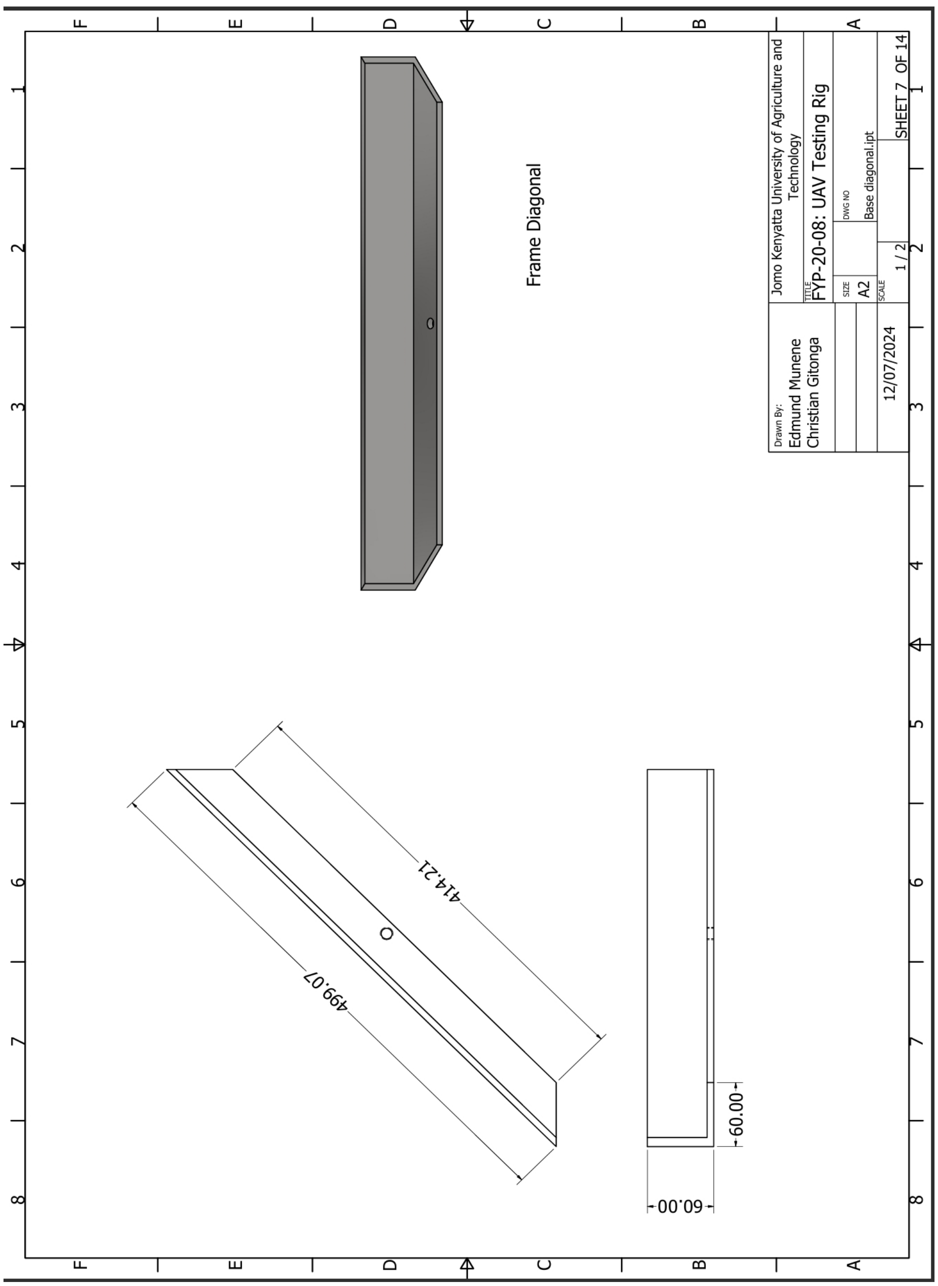



Figure C.5: Frame diagonal

### C.2.3 Frame Stand

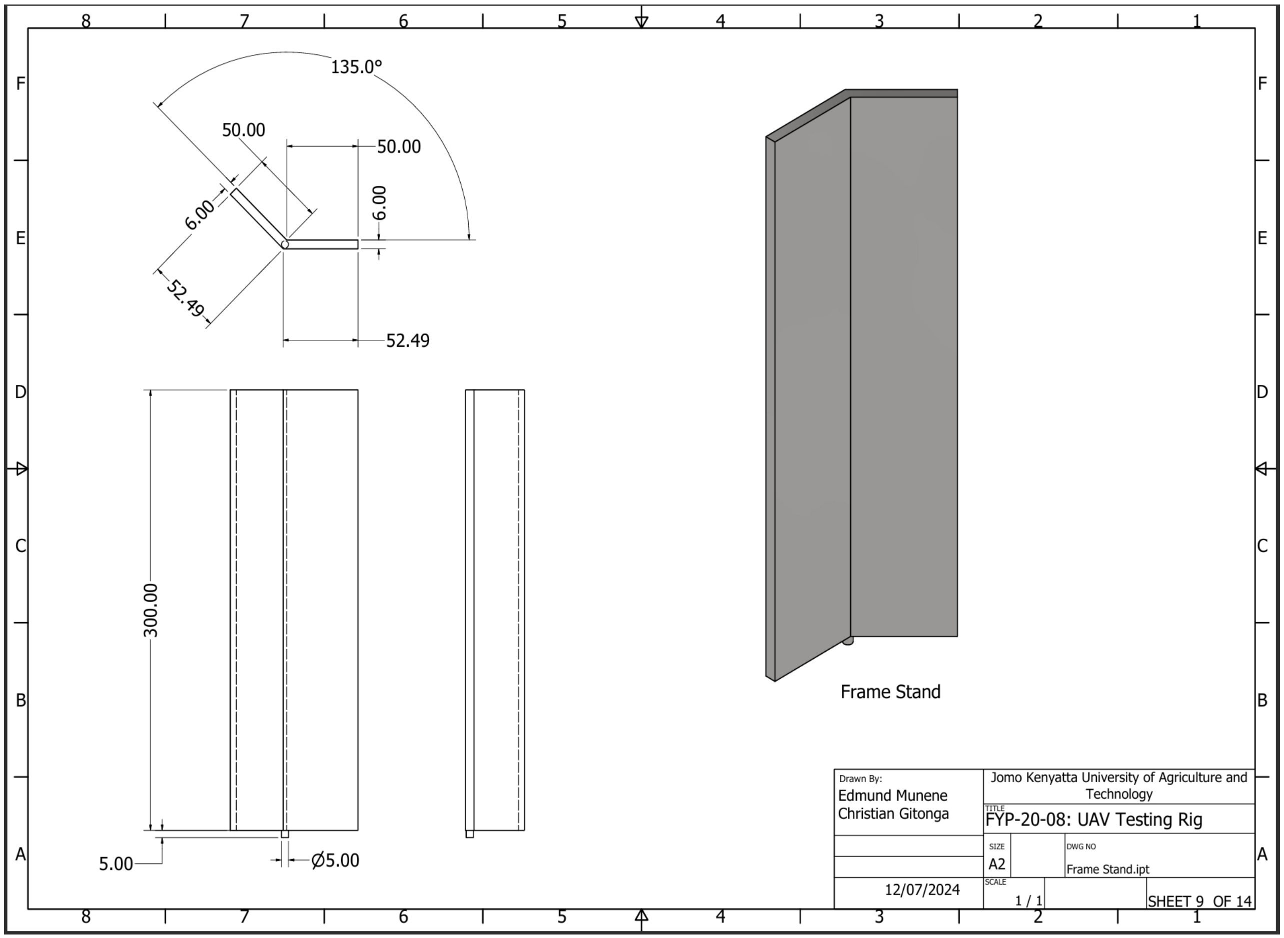



Figure C.6: Frame stand

### C.2.4 Support Base

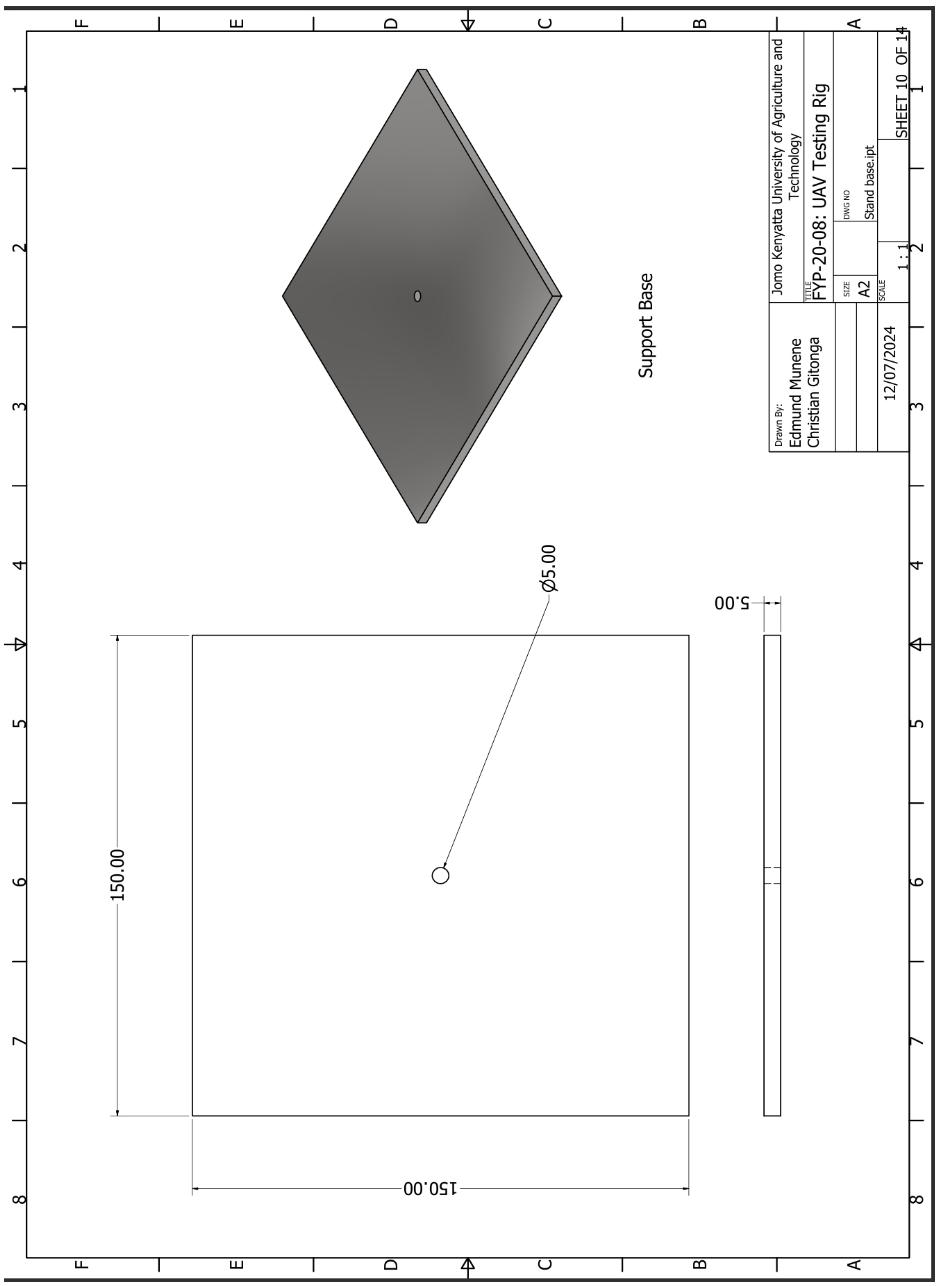



Figure C.7: Support base

## C.3 Plate Assembly

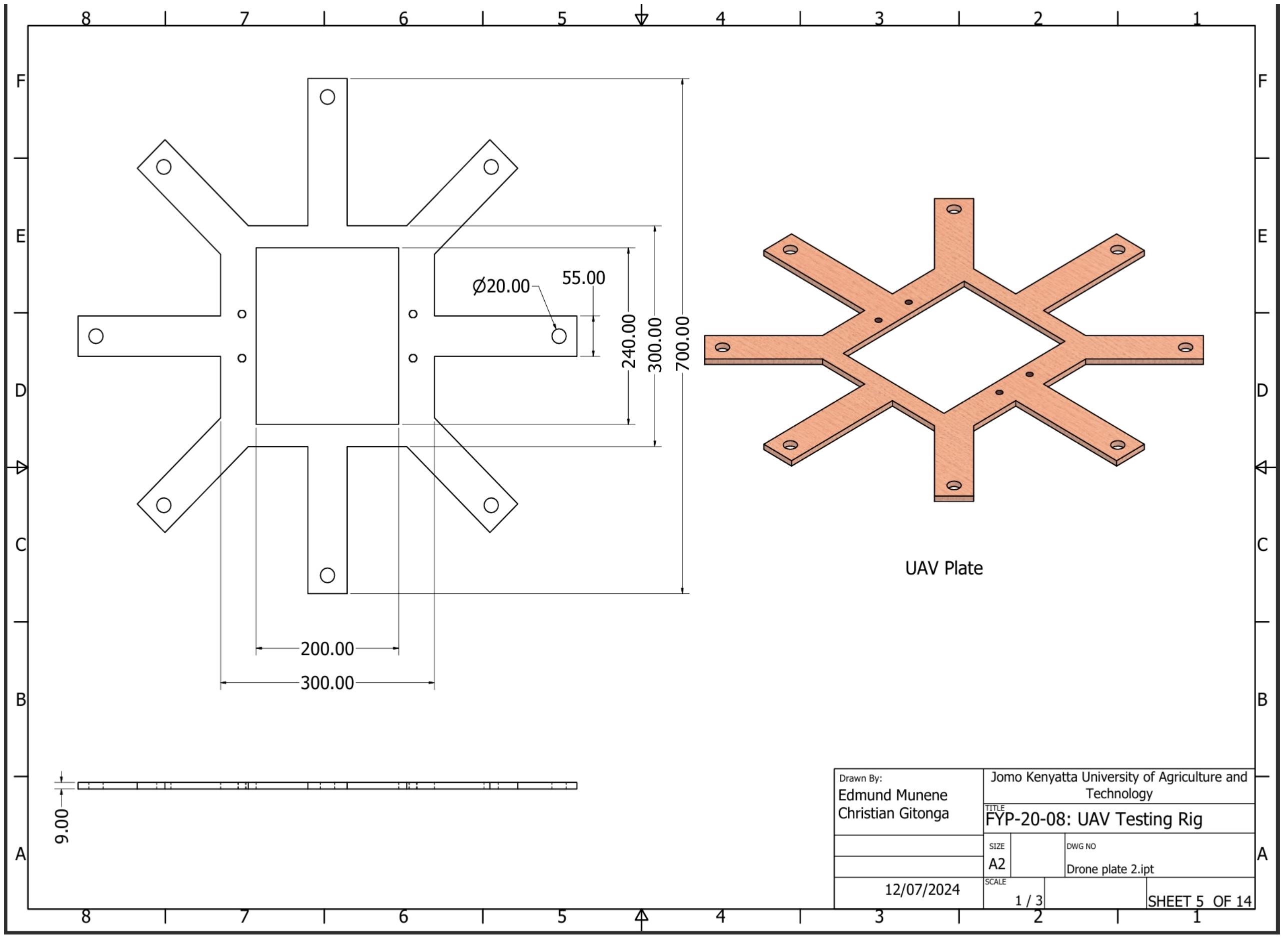



Figure C.8: Plate assembly

### C.3.1 UAV plate

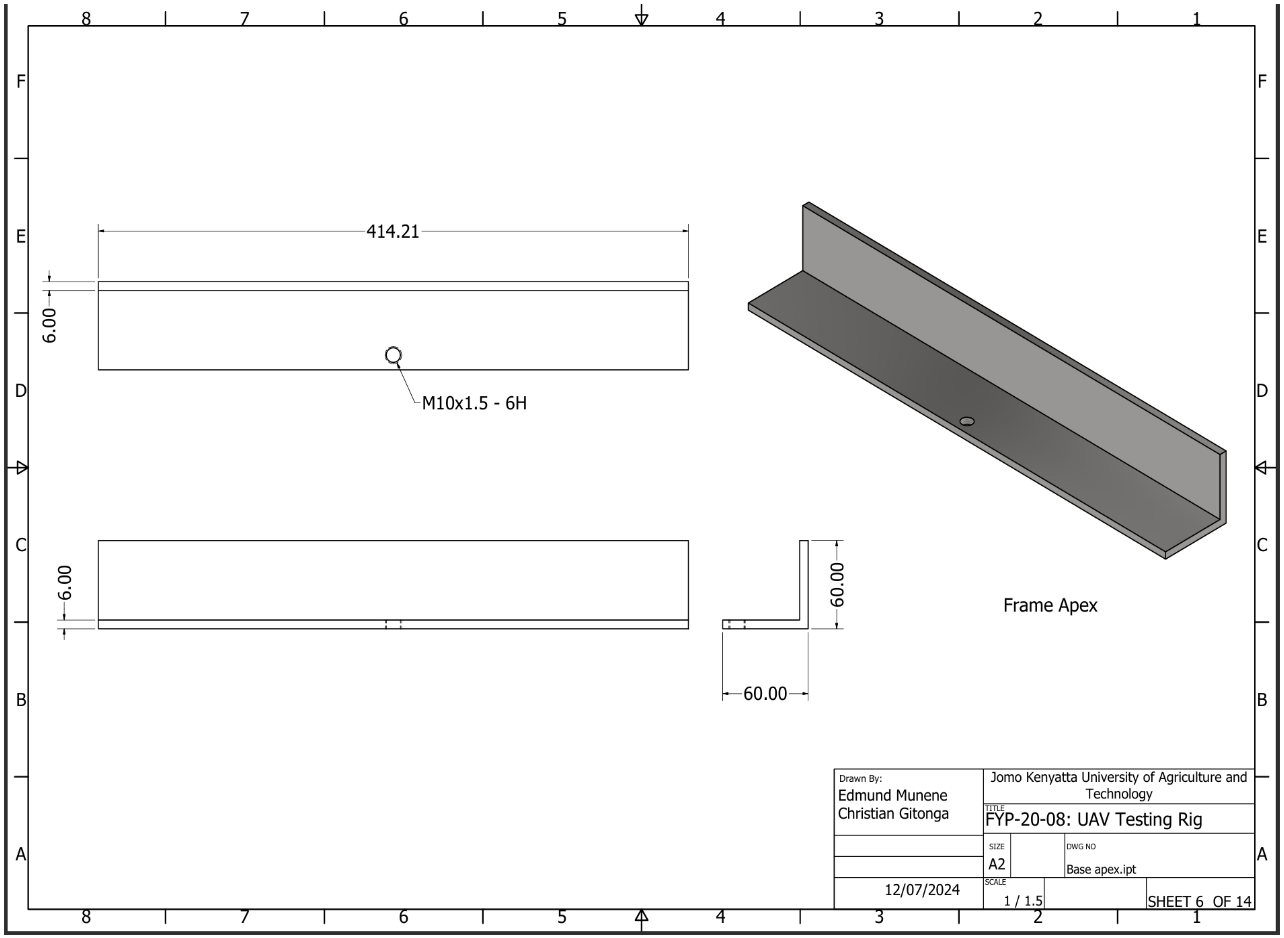



Figure C.9: UAV plate

# D Electrical Circuit

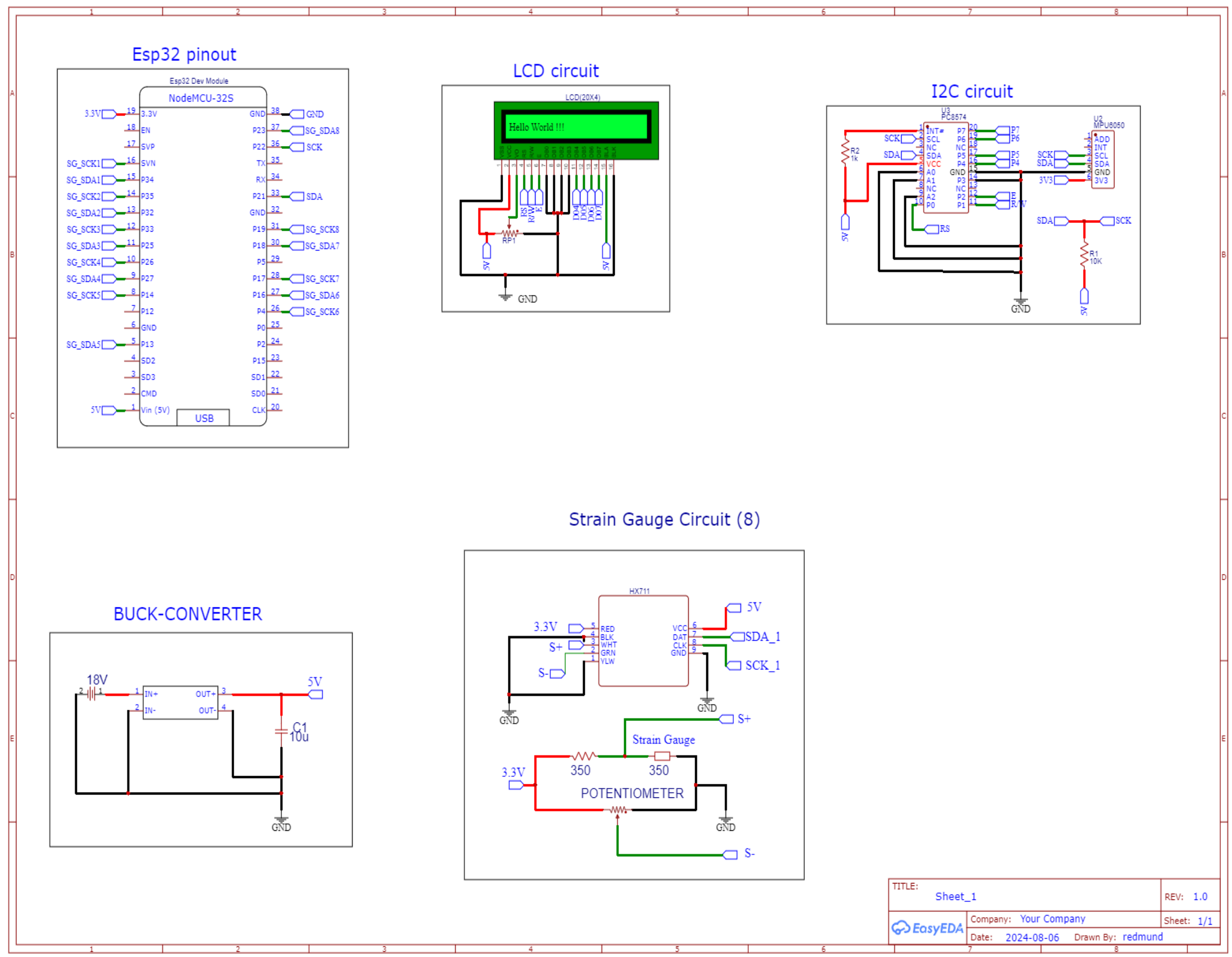



Figure D.1: Electrical Circuit